# UNIT-3

## Exploratory Data Analytics

### What is Exploratory Data Analysis?

Exploratory data analysis is one of the basic and essential steps of a data science project. A data scientist involves almost 70% of his work in doing the EDA of the dataset. In this article, we will discuss what is **Exploratory Data Analysis (EDA)** and the steps to perform EDA.

### What is Exploratory Data Analysis (EDA)?

Exploratory Data Analysis (EDA) is a crucial initial step in data science projects. It involves analyzing and visualizing data to understand its key characteristics, uncover patterns, and identify relationships between variables refers to the method of studying and exploring record sets to apprehend their predominant traits, discover patterns, locate outliers, and identify relationships between variables. EDA is normally carried out as a preliminary step before undertaking extra formal statistical analyses or modeling.

**Key aspects of EDA include:**

- **Distribution of Data**: Examining the distribution of data points to understand their range, central tendencies (mean, median), and dispersion (variance, standard deviation).
- **Graphical Representations**: Utilizing charts such as histograms, box plots, scatter plots, and bar charts to visualize relationships within the data and distributions of variables.
- **Outlier Detection**: Identifying unusual values that deviate from other data points. Outliers can influence statistical analyses and might indicate data entry errors or unique cases.
- **Correlation Analysis**: Checking the relationships between variables to understand how they might affect each other. This includes computing correlation coefficients and creating correlation matrices.
- **Handling Missing Values**: Detecting and deciding how to address missing data points, whether by imputation or removal, depending on their impact and the amount of missing data.
- **Summary Statistics:** Calculating key statistics that provide insight into data trends and nuances.
- **Testing Assumptions**: Many statistical tests and models assume the data meet certain conditions (like normality or homoscedasticity). EDA helps verify these assumptions.

### Why Exploratory Data Analysis is Important?

Exploratory Data Analysis (EDA) is important for several reasons, especially in the context of data science and statistical modeling. Here are some of the key reasons why EDA is a critical step in the data analysis process:

1. **Understanding Data Structures**: EDA helps in getting familiar with the dataset, understanding the number of features, the type of data in each feature, and the distribution of data points. This understanding is crucial for selecting appropriate analysis or prediction techniques.
2. **Identifying Patterns and Relationships**: Through visualizations and statistical summaries, EDA can reveal hidden patterns and intrinsic relationships between variables. These insights can guide further analysis and enable more effective feature engineering and model building.
3. **Detecting Anomalies and Outliers**: EDA is essential for identifying errors or unusual data points that may adversely affect the results of your analysis. Detecting these early can prevent costly mistakes in predictive modeling and analysis.
4. **Testing Assumptions**: Many statistical models assume that data follow a certain distribution or that variables are independent. EDA involves checking these assumptions. If the assumptions do not hold, the conclusions drawn from the model could be invalid.
5. **Informing Feature Selection and Engineering**: Insights gained from EDA can inform which features are most relevant to include in a model and how to transform them (scaling, encoding) to improve model performance.
6. **Optimizing Model Design**: By understanding the data's characteristics, analysts can choose appropriate modeling techniques, decide on the complexity of the model, and better tune model parameters.
7. **Facilitating Data Cleaning**: EDA helps in spotting missing values and errors in the data, which are critical to address before further analysis to improve data quality and integrity.

8. **Enhancing Communication**: Visual and statistical summaries from EDA can make it easier to communicate findings and convince others of the validity of your conclusions, particularly when explaining data-driven insights to stakeholders without technical backgrounds.

**Types of Exploratory Data Analysis**

EDA, or Exploratory Data Analysis, refers back to the method of analyzing and analyzing information units to uncover styles, pick out relationships, and gain insights. There are various sorts of EDA strategies that can be hired relying on the nature of the records and the desires of the evaluation. Depending on the number of columns we are analyzing we can divide EDA into three types: Univariate, bivariate and multivariate.

1. Univariate Analysis

Univariate analysis focuses on a single variable to understand its internal structure. It is primarily concerned with describing the data and finding patterns existing in a single feature. This sort of evaluation makes a speciality of analyzing character variables inside the records set. It involves summarizing and visualizing a unmarried variable at a time to understand its distribution, relevant tendency, unfold, and different applicable records. Common techniques include:

- **Histograms**: Used to visualize the distribution of a variable.
- **Box plots**: Useful for detecting outliers and understanding the spread and skewness of the data.
- **Bar charts**: Employed for categorical data to show the frequency of each category.
- **Summary statistics**: Calculations like mean, median, mode, variance, and standard deviation that describe the central tendency and dispersion of the data.
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2. Bivariate Analysis

Bivariate evaluation involves exploring the connection between variables. It enables find associations, correlations, and dependencies between pairs of variables. Bivariate analysis is a crucial form of exploratory data analysis that examines the relationship between two variables. Some key techniques used in bivariate analysis:

- **Scatter Plots:** These are one of the most common tools used in bivariate analysis. A scatter plot helps visualize the relationship between two continuous variables.
- **Correlation Coefficient**: This statistical measure (often Pearson's correlation coefficient for linear relationships) quantifies the degree to which two variables are related.
- **Cross-tabulation**: Also known as contingency tables, cross-tabulation is used to analyze the relationship between two categorical variables. It shows the frequency distribution of categories of one variable in rows and the other in columns, which helps in understanding the relationship between the two variables.
- **Line Graphs**: In the context of time series data, line graphs can be used to compare two variables over time. This helps in identifying trends, cycles, or patterns that emerge in the interaction of the variables over the specified period.

- **Covariance**: Covariance is a measure used to determine how much two random variables change together. However, it is sensitive to the scale of the variables, so it's often supplemented by the correlation coefficient for a more standardized assessment of the relationship.

3. Multivariate Analysis

Multivariate analysis examines the relationships between two or more variables in the dataset. It aims to understand how variables interact with one another, which is crucial for most statistical modeling techniques. Techniques include:

- **Pair plots**: Visualize relationships across several variables simultaneously to capture a comprehensive view of potential interactions.
- **Principal Component Analysis (PCA)**: A dimensionality reduction technique used to reduce the dimensionality of large datasets, while preserving as much variance as possible.

Specialized EDA Techniques

In addition to univariate and multivariate analysis, there are specialized EDA techniques tailored for specific types of data or analysis needs:

- **Spatial Analysis**: For geographical data, using maps and spatial plotting to understand the geographical distribution of variables.
- **Text Analysis**: Involves techniques like word clouds, frequency distributions, and sentiment analysis to explore text data.
- **Time Series Analysis:** This type of analysis is mainly applied to statistics sets that have a temporal component. Time collection evaluation entails inspecting and modeling styles, traits, and seasonality inside the statistics through the years. Techniques like line plots, autocorrelation analysis, transferring averages, and ARIMA (AutoRegressive Integrated Moving Average) fashions are generally utilized in time series analysis.

**Tools for Performing Exploratory Data Analysis**

Exploratory Data Analysis (EDA) can be effectively performed using a variety of tools and software, each offering unique features suitable for handling different types of data and analysis requirements.

1. Python Libraries

- **Pandas**: Provides extensive functions for data manipulation and analysis, including data structure handling and time series functionality.
- **Matplotlib**: A plotting library for creating static, interactive, and animated visualizations in Python.
- **Seaborn**: Built on top of Matplotlib, it provides a high-level interface for drawing attractive and informative statistical graphics.
- **Plotly**: An interactive graphing library for making interactive plots and offers more sophisticated visualization capabilities.

2. R Packages

- **ggplot2**: Part of the tidyverse, it's a powerful tool for making complex plots from data in a data frame.
- **dplyr**: A grammar of data manipulation, providing a consistent set of verbs that help you solve the most common data manipulation challenges.
- **tidyr**: Helps to tidy your data. Tidying your data means storing it in a consistent form that matches the semantics of the dataset with the way it is stored.

**Mean, Median and Mode: - Mean, Median, and Mode** are measures of the central tendency. These values are used to define the various parameters of the given data set. The measure of central tendency (Mean, Median, and Mode) gives useful insights about the data studied, these are used to study any type of data such as the average salary of employees in an organization, the median age of any class, the number of people who plays cricket in a sports club, etc.

**Measures of Central Tendency**

Measure of central tendency is the representation of various values of the given data set. There are various measures of central tendency and the most important three measures of central tendency are:

- **Mean**
- **Median**

- **Mode**

**What are Mean, Median, and Mode?**

Mean, median, and mode are measures of central tendency used in statistics to summarize a set of data.

**Mean ($\bar{x}$ or $\mu$):** The mean, or arithmetic average, is calculated by summing all the values in a dataset and dividing by the total number of values. It's sensitive to outliers and is commonly used when the data is symmetrically distributed.

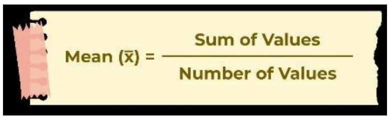
**Median (M):** The median is the middle value when the dataset is arranged in ascending or descending order. If there's an even number of values, it's the average of the two middle values. The median is robust to outliers and is often used when the data is skewed.

**Mode (Z):** The mode is the value that occurs most frequently in the dataset. Unlike the mean and median, the mode can be applied to both numerical and categorical data. It's useful for identifying the most common value in a dataset.

**What is Mean?**

**Mean** is the sum of all the values in the data set divided by the number of values in the data set. It is also called the Arithmetic Average. Mean is denoted as $\bar{x}$ and is read as **x bar**.

The formula to calculate the mean is:




Mean Symbol

The symbol used to represent the mean, or arithmetic average, of a dataset is typically the Greek letter "$\mu$" (mu) when referring to the population mean, and "$\bar{x}$" (x-bar) when referring to the sample mean.

- Population Mean: **$\mu$ (mu)**
- Sample Mean: **$\bar{x}$ (x-bar)**

**These symbols are commonly used in statistical notation to represent the average value of a set of data points.**

**Mean Formula**

The formula to calculate the mean is:

$$\text{Mean } (\bar{x}) = \text{Sum of Values / Number of Values}$$

If x1, x2, x3,……, xn are the values of a data set then the mean is calculated as:

$$\bar{x} = (x1 + x2 + x3 + . . . + xn) / n$$

**Example: Find the mean of data sets 10, 30, 40, 20, and 50.**

**Solution:**

Mean of the data 10, 30, 40, 20, 50 is

Mean = (sum of all values) / (number of values)

Mean = (10 + 30 + 40 + 20+ 50) / 5 = 30

Mean of Grouped Data

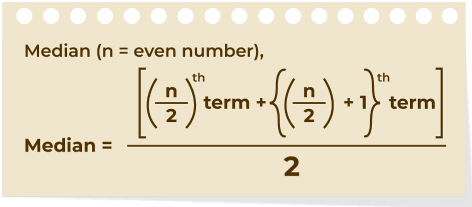
Mean for the grouped data can be calculated by using various methods. The most common methods used are discussed in the table below:

| Direct Method | Assumed Mean Method | Step Deviation Method |
|---|---|---|
| $\bar{x} = \sum fixi / \sum fi$<br>Where,<br>**$\sum$fi** is the sum of all frequencies | $\bar{x} = a + \sum fixi / \sum fi$<br>Where,<br>**a** is Assumed mean<br>**di** is equal to xi – a<br>**$\sum$fi** the sum of all frequencies | $\bar{x} = a + h\sum fixi / \sum fi$<br>Where,<br>a is Assumed mean<br>**ui** = (xi – a)/h<br>**h** is Class size<br>**$\sum$fi** the sum of all frequencies |

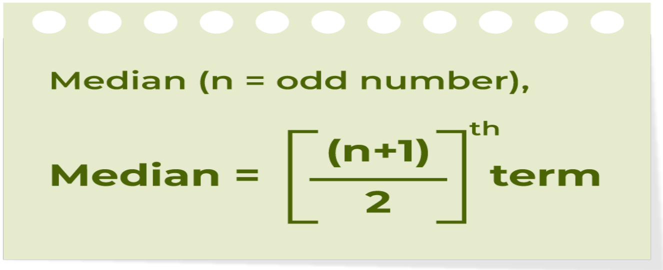
**What is Median?**

A Median is a middle value for sorted data. The sorting of the data can be done either in ascending order or descending order. A median divides the data into two equal halves.

The formula to calculate the median of the number of terms if the number of terms is even is shown in the image below:




The formula to calculate the median of the number of terms if the number of terms is odd is shown in the image below:




Median Symbol

The letter **"M"** is commonly used to represent the median of a dataset, whether it's for a population or a sample. This notation simplifies the representation of statistical concepts and calculations, making it easier to understand and apply in various contexts. Therefore, in Indian statistical practice, **"M" is widely accepted and understood as the symbol for the median.**

**Median Formula**

The formula for the median is:

If the number of values (n value) in the data set is odd then the formula to calculate the median is:

***Median = [(n + 1)/2]th term***

If the number of values (n value) in the data set is even then the formula to calculate the median is:

***Median = [(n/2)th term + {(n/2) + 1}th term] / 2***

**Example: Find the median of given data set 30, 40, 10, 20, and 50.**

**Solution:**

*Median of the data 30, 40, 10, 20, 50 is,*

***Step 1:*** *Order the given data in ascending order as:*

*10, 20, 30, 40, 50*

***Step 2:*** *Check n (number of terms of data set) is even or odd and find the median of the data with respective 'n' value.*

***Step 3:*** *Here, n = 5 (odd)*

*Median = [(n + 1)/2]th term*

*Median = [(5 + 1)/2]th term*

*= 30*

Median of Grouped Data

The median of the grouped data median is calculated using the formula,

***Median = l + [(n/2 – cf) / f]×h***

where

- **l** is lower limit of median class
- **n** is number of observations
- **f** is frequency of median class
- **h** is class size
- **cf** is cumulative frequency of class preceding the median class.

**What is Mode?**

A mode is the most frequent value or item of the data set. A data set can generally have one or more than one mode value. If the data set has one mode then it is called "Uni-modal". Similarly, If the data

set contains 2 modes then it is called “Bimodal” and if the data set contains 3 modes then it is known as “Trimodal”. If the data set consists of more than one mode then it is known as “multi-modal”(can be bimodal or trimodal). There is no mode for a data set if every number appears only once.
The formula to calculate the mode is shown in the image below:


Symbol of Mode
In statistical notation, the symbol **“Z”** is commonly used to represent the mode of a dataset. It indicates the value or values that occur most frequently within the dataset. This symbol is widely utilised in statistical discourse to signify the mode, enhancing clarity and precision in statistical discussions and analyses.

**Mode Formula**

$$Mode = Highest\ Frequency\ Term$$

**Example: Find the mode of the given data set 1, 2, 2, 2, 3, 3, 4, 5.**

**Solution:**

*Given set is {1, 2, 2, 2, 3, 3, 4, 5}*
*As the above data set is arranged in ascending order.*
*By observing the above data set we can say that,*
*Mode = 2*
*As, it has highest frequency (3)*

Mode of Grouped Data
The mode of the grouped data is calculated using the formula:

$$Mode = l + [(f1 + f0) / (2f1 – f0 – f2)] \times h$$

where,

- **f1** is the frequency of the modal class,
- **f0** is the frequency of the class preceding the modal class,
- **f2** is the frequency of the class succeeding the modal class,
- **h** is the size of class intervals, and
- **l** is the lower limit of modal class.

***Read More about Mode of Grouped Data.***

**Relation between Mean Median Mode**

For any group of data, the relation between the three central tendencies mean, median, and mode is shown in the image below:

$$Mode = 3\ Median – 2\ Mean$$


*Mode = 3 Median – 2 Mean*

**Mean, Median and Mode:** Another name for this relationship is an empirical relationship. When we know the other two measures for a given set of data, this is used to find one of the measures. The LHS and RHS can be switched to rewrite this relationship in various ways.

**What is Range?**

In a given data set the difference between the largest value and the smallest value of the data set is called the range of data set. For example, if height(in cm) of 10 students in a class are given in ascending order, 160, 161, 167, 169, 170, 172, 174, 175, 177, and 181 respectively. Then range of data set is (181 – 160) = 21 cm.

Range of Data

**Range** is the difference between the highest value and the lowest value. It is a way to understand how the numbers are spread in a data set. The range of any data set is easily calculated by using the formula given in the image below:


*Formula to Find Range*

Range Formula

The formula to find the Range is:

***Range = Highest value – Lowest Value***

**Example: Find the range of the given data set 12, 19, 6, 2, 15, 4.**

**Solution:**

*Given set is {12, 19, 6, 2, 15, 4}*

*Here,*

*Lowest Value = 2*

*Highest Value = 19*

*Range = 19 − 2 = 17*

**Difference Between Mean and Median**

The key differences between mean and median are listed in the following table:

| Aspect | Mean | Median |
|---|---|---|
| **Definition** | The sum of all values divided by the count | The middle value of a sorted dataset |
| **Calculation** | Mean = Sum of all values/Count | Median is the middle value when the data is arranged in ascending or descending order |
| **Sensitivity to Outliers** | Can be highly influenced by extreme values in the dataset | Less sensitive to extreme values, outliers have minimal impact |
| **Use Cases** | Commonly used in statistical analysis and mathematics | Useful when extreme values skew the data or when the distribution is not symmetric |

**Let's see the following example to understand the difference.**

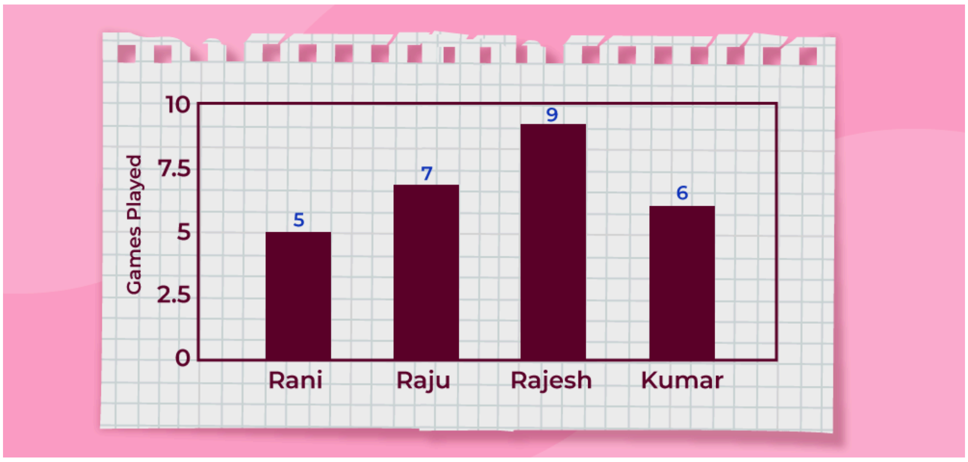
*Difference between Mean and Median is understood by the following example. In a school, there are 8 teachers whose salaries are 20000 rupees, a principal with a salary of 35000, find their mean salary and median salary.*

*Mean = (20000+20000+20000+20000+20000+20000+20000+20000+35000)/9 = 195000/9 = 21666.67*
*Therefore, the* ***mean salary is ₹21,666.67.***
*For median, in ascending order: 20000, 20000, 20000, 20000, 20000, 20000, 20000, 20000, 35000.*
*n = 9,*
*Thus, (9 + 1)/2 = 5*
*Thus, the* ***median is 5th observation.***
***Median = 20000***
*Therefore, the* ***median is ₹20,000.***

**Differences between Mean, Median and Mode**
Mean, median, and mode are measures of central tendency in statistics.

| Feature | Mean | Median | Mode |
|---|---|---|---|
| **Definition** | Mean is the average of all values. | Median is the middle value when data is sorted. | Mode is the most frequently occurring value in the dataset. |
| **Sensitivity** | Mean is sensitive to outliers. | Median is not sensitive to outliers . | Mode is not sensitive to outliers. |
| **Calculation** | Calculated by adding up all values of a dataset and dividing them by the total number of values in dataset. | Calculated by finding the middle value in a list of data. | Calculated by finding which value occurs more number of times in a dataset. |
| **Representation** | Value of mean may or may not be in dataset. | Value of median is always a value from the dataset. | Value of mode is also always a value from the dataset. |

**Question 1:** Study the bar graph given below and find the mean, median, and mode of the given data set.




**Solution:**

***Mean = (sum of all data values) / (number of values)***

*Mean = (5 + 7 + 9 + 6) / 4*

*= 27 / 2*

*= 6.75*

*Order the given data in ascending order as: 5, 6, 7, 9*

*Here, n = 4 (which is even)*

***Median = [(n/2)th term + {(n/2) + 1}th term] / 2***

*Median = (6 + 7) / 2*

*= 6.5*

***Mode = Most frequent value***

*= 9 (highest value)*

***Range = Highest value – Lowest value***

*Range = 9 – 5*

*= 4*

Question 2: Find the mean, median, mode, and range for the given data

**190, 153, 168, 179, 194, 153, 165, 187, 190, 170, 165, 189, 185, 153, 147, 161, 127, 180**

**Solution:**

***For Mean:***

*190, 153, 168, 179, 194, 153, 165, 187, 190, 170, 165, 189, 185, 153, 147, 161, 127, 180*

*Number of observations = 18*

*Mean = (Sum of observations) / (Number of observations)*

*= (190+153+168+179+194+153+165+187+190+170+165+189+185+153+147 +161+127+180) / 18*

*= 2871/18*

*= 159.5*

*Therefore, the mean is 159.5*

***For Median:***

*The ascending order of given observations is,*

*127, 147, 153, 153, 153, 161, 165, 165, 168, 170, 179, 180, 185, 187, 189, 190, 190, 194*

*Here, n = 18*

*Median = 1/2 [(n/2) + (n/2 + 1)]th observation*

*= 1/2 [9 + 10]th observation*

*= 1/2 (168 + 170)*

*= 338/2*

*= 169*

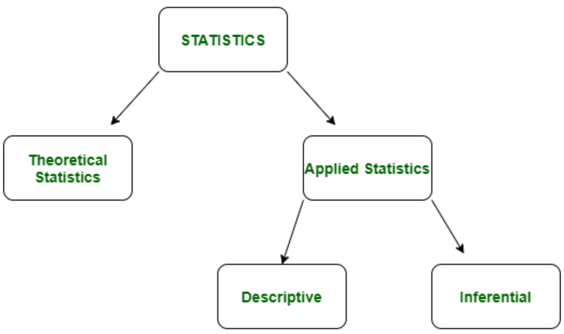
*Thus, the median is 169*

***For Mode:***

*The number with the highest frequency = 153*
*Thus, mode = 53*
***For Range:***
*Range = Highest value – Lowest value*
*= 194 – 127*
*= 67*
Question 3: Find the Median of the data 25, 12, 5, 24, 15, 22, 23, 25
**Solution:**
*25, 12, 5, 24, 15, 22, 23, 25*
***Step 1:*** *Order the given data in ascending order as:*
*5, 12, 15, 22, 23, 24, 25, 25*
***Step 2:*** *Check n (number of terms of data set) is even or odd and find the median of the data with respective 'n' value.*
***Step 3:*** *Here, n = 8 (even) then,*
*Median = [(n/2)th term + {(n/2) + 1)th term] / 2*
*Median = [(8/2)th term + {(8/2) + 1}th term] / 2*
*= (22+23) / 2*
*= 22.5*
Question 4: Find the mode of given data 15, 42, 65, 65, 95.
**Solution:**
*Given data set 15, 42, 65, 65, 95*
*The number with highest frequency = 65*
*Mode = 65*

**Difference between Descriptive and Inferential statistics**
**Statistics is a vital discipline that empowers us to make sense of data by providing tools for collection, analysis, interpretation, and presentation.** In every field, from engineering to social sciences, understanding data is **crucial for making informed decisions and drawing accurate conclusions**. This understanding is facilitated by two key branches of statistics: descriptive and inferential.

**What is Statistics?**
Statistics is a **branch of mathematics dealing with the collection, analysis, interpretation, and presentation of masses of numerical data**. It is basically a collection of quantitative data.
Statistics is a fundamental branch of mathematics that involves the collection, analysis, interpretation, presentation, and organization of data. statistics is **divided into two main branches: descriptive statistics and inferential statistics**. These two branches serve different purposes and are used in various fields, including engineering, social sciences, business, and healthcare. This article explores the definitions, characteristics, and applications of both descriptive and inferential statistics.
**Types of Statistics:**




**Difference between Descriptive and Inferential statistics**

| **Descriptive Statistics** | **Inferential Statistics** |
|---|---|
| It **gives information about raw data** which describes the data in some manner. | It **makes inferences about the population** using data drawn from the population. |
| It **helps in organizing, analyzing, and to present data in a meaningful manner**. | It **allows us to compare data, and make hypotheses and predictions.** |
| It is used to describe a situation. | It is **used to explain the chance of occurrence of an event**. |
| It explains already known data and is limited to a sample or population having a small size. | It attempts to reach the conclusion about the population. |
| It **can be achieved with the help of charts, graphs, tables,** etc. | It can be achieved by probability. |

**Descriptive Statistics**

**Descriptive statistics** is a term given to the **analysis of data that helps to describe, show and summarize data in a meaningful way**. It is a simple way to describe our data. **Descriptive statistics is very important to present our raw data in effective/meaningful way using numerical calculations or graphs or tables**. This type of statistics is applied to already known data.

**Descriptive statistics involves summarizing and organizing data to describe the main features of a dataset**. It provides simple summaries about the sample and the measures. Descriptive statistics is primarily concerned with the presentation of data in a meaningful way, which includes graphical representation and numerical analysis.

Uses cases of Descriptive Statistics

*Measures of Central Tendency*

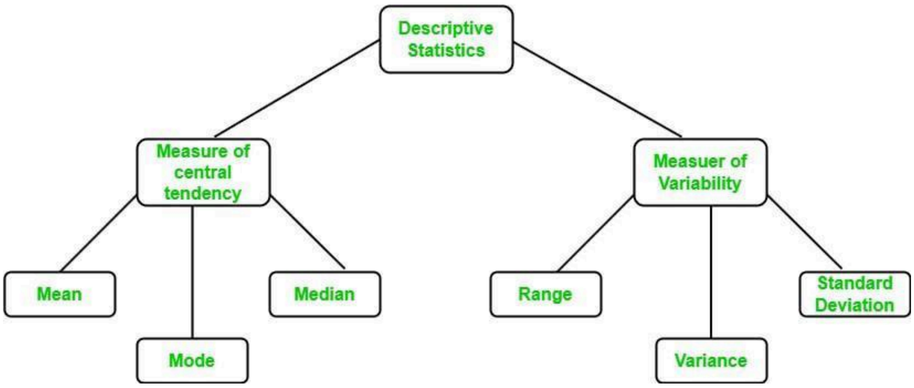
- **Mean**: The average of all data points.
- **Mode**: The most frequently occurring value in a dataset.
- **Median**: The middle value that separates the higher half from the lower half of the data.

*Graphical Representation*

- **Histograms**: Bar graphs representing the frequency distribution of a dataset.
- **Pie Charts**: Circular charts divided into sectors representing relative frequencies.
- **Box Plots**: Graphical depiction of data through their quartiles.

*Measures of Dispersion*

- **Range**: The **difference between the maximum and minimum values.**
- **Variance**: The measure of **how data points differ from the mean**.
- **Standard Deviation**: The square root of the variance, **representing the average distance from the mean.**



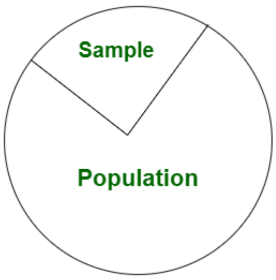

*Descriptive Statistics*

**Applications of Descriptive Statistics**

- **Business Analysis**: Summarizing sales data to identify trends and make informed business decisions.
- **Healthcare**: Analyzing patient data to understand the distribution of health outcomes.
- **Engineering**: Monitoring manufacturing processes through quality control charts to ensure consistency.

**2. Inferential Statistics**

**[Inferential statistics](#) is used to make predictions by taking any group of data in which you are interested**. It can be defined as a random sample of data taken from a population to describe and make [inferences](#) about the population. Any **group of data that includes all the data you are interested in is known as population.** It basically **allows you to make predictions by taking a small sample instead of working on the whole population.**




**Uses cases of Inferential Statistics**

Estimation

- **Point Estimation**: **Provides a single value estimate of a population parameter** (e.g., sample mean as an estimate of population mean).
- **Interval Estimation**: Provides a range of values within which the population parameter is expected to lie (e.g., confidence intervals).

Hypothesis Testing

- **Null Hypothesis (H0)**: A **statement of no effect or no difference**, which researchers aim to test against.
- **Alternative Hypothesis (H1)**: A statement indicating the presence of an effect or difference.
- **p-value**: The probability of observing the test results under the null hypothesis.
- **Significance Level (α)**: The threshold for rejecting the [null hypothesis](#), commonly set at 0.05.

Regression Analysis

- **Simple Linear Regression**: Analyzing the relationship between two [continuous variables.](#)
- **Multiple Regression**: Examining the relationship between one dependent variable and multiple independent variables.

**Applications of Inferential Statistics**

- **Market Research**: Making predictions about consumer behavior based on survey samples.
- **Clinical Trials**: Drawing conclusions about the effectiveness of new treatments from sample data.
- **Engineering**: Predicting product performance and reliability through sample testing and analysis.

**Conclusion**

**Descriptive and inferential statistics are essential tools in the field of statistics, each serving distinct but complementary purposes.** Descriptive statistics focuses on summarizing and presenting data to highlight its main features, while inferential statistics aims to make predictions and generalizations about a population based on sample data. **Understanding and applying these two branches of statistics enables researchers, analysts, and engineers to make informed decisions, draw meaningful conclusions**, and advance knowledge in their respective fields.

**Standard Deviation – Formula, Examples & How to Calculate**

**Standard Deviation** is the measure of the dispersion of statistics. The standard deviation formula is used to find the deviation of the data value from the mean value i.e. it is used to find the dispersion of all the values in a data set to the mean value. There are different standard deviation formulas to calculate the standard deviation of a random variable.

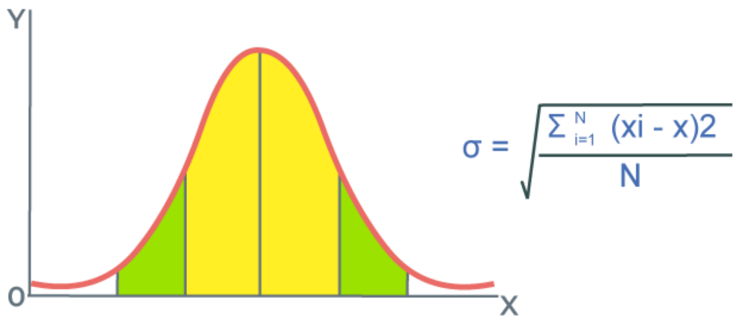
In this article, we will learn about **what is standard deviation, the standard deviation formulas, how to calculate standard deviation, and examples of standard deviation in detail.**

**What is Standard Deviation?**

Standard Deviation is defined as the degree of dispersion of the data point to the mean value of the data point. It tells us how the value of the data points varies to the mean value of the data point and it tells us about the variation of the data point in the sample of the data.

Standard Deviation of a given sample of data set is also defined as the square root of the variance of the data set. Mean Deviation of the n values (say x1, x2, x3, …, xn) is calculated by taking the sum of the squares of the difference of each value from the mean, i.e.

***Mean Deviation = 1/n∑in (xi – x̄)2***




Mean Deviation **is used to tell us about the scatter of the data.** The lower degree of deviation tells us that the observations xi are close to the mean value and the depression is low. In contrast, the higher degree of deviation tells us that the observations xi are far from the mean value and the dispersion is high.

**Standard Deviation Definition**

Standard deviation is a measure used in statistics to understand how the data points in a set are spread out from the mean value. It indicates the extent of the data's variation and shows how far individual data points deviate from the average.

**Standard Deviation Formula**

Standard deviation is used to measure the spread of the statistical data. It tells us about how the statistical data is spread out. **Formula to Calculate Standard Deviation is used to find the deviation of all the data sets from its mean position**. You may have questions that standard deviation how to calculate or **how to calculate a standard deviation**. There are two standard deviation formulas that are used to find the Standard Deviation of any given data set. They are,

- *Population Standard Deviation Formula*
- *Standard Deviation Formula Sample*

Formula for Standard Deviation of Sample Data is,

*s=∑i=1n(xi–x¯)2n−1* ***s*=*n*−1∑*i*=1*n*(*xi*–*x*¯)2**

where,

- **s** is Population Standard Deviation
- **xi** is ith observation
- **x̄** is Sample Mean
- **N** is Number of Observations

Standard Deviation Formula of Population Data is,

*σ=∑i=1N(xi–μ)2N σ=****N****∑****i****=****1N****(****xi****–μ)****2***

where,

- **σ** is Population Standard Deviation
- **xi** is ith Observation
- **μ** is Population Mean
- **N** is Number of Observations

**It is evident to note that both formulas look the same and have only slide changes in their denominator.** Denominator in case of the sample is **n-1** but in case of the **population is N.** Initially the denominator in the **sample standard deviation** formula has **"n"** in its denominator but the result from this formula was not appropriate. So, a correction was made and **the n is replaced with n-1 this correction is called Bessel's correction** which in turn produced the most appropriate results.

Formula for Calculating Standard Deviation

Formula used for calculating Standard Deviation is discussed in the image below,

| **Population** | **Sample** |
|---|---|
| $\sigma = \sqrt{\dfrac{\Sigma(x_i-\mu)^2}{n}}$ | $S = \sqrt{\dfrac{\Sigma(x_i-\overline{x})^2}{n-1}}$ |
| **μ - Population Average**<br>**xi - Individual Population Value**<br>**n - Total Number of Population** | **x - Sample Average**<br>**$x_i$ - Individual Population Value**<br>**n - Total Number of Sample** |

**How to Calculate Standard Deviation?**

Generally, when we talk about standard deviation we talk about **population standard deviation**. The steps to calculate standard deviation of a given set of values are as follows,

**Step 1: Calculate mean of observation using the formula**

***(Mean = Sum of Observations/Number of Observations)***

**Step 2: Calculate squared differences of data values from the mean.**

*(Data Value – Mean)2*

**Step 3: Calculate average of squared differences.**

*(Variance = Sum of Squared Differences / Number of Observations)*

**Step 4: Calculate square root of variance this gives the Standard Deviation.**

***(Standard Deviation = √Variance)***

**What is Variance?**

Variance basically tells us how spread out a set of data is. If all the data points are the same, the variance is zero. **Any non-zero variance is considered positive**. **Low variance means the data points are close to the average (or mean) and to each other. High variance means the data points are spread out from the average and from each other.** In simple terms, variance is the average of how far each data point is from the mean, squared.

Difference between Variance and Standard Deviation

| Aspect | Variance | Deviation (Standard Deviation) |
|---|---|---|
| **Definition** | Measure of spread in a dataset. | Measure of average distance from the mean. |
| **Calculation** | Average of squared differences from the mean. | Square root of the variance. |
| **Symbol** | σ^2 (sigma squared) | σ (sigma) |
| **Interpretation** | Indicates the average squared deviation of data points from the mean. | Indicates the average distance of data points from the mean. |

**Variance Formula**

The formula to calculate the variance of a dataset is as follows:

***Variance (σ2) = Σ [(x – μ)2] / N***

where:

- **Σ** denotes Summation (adding up)
- **x** represents Each Individual Data Point
- **μ** is the Mean (Average) of Dataset
- **N** is the Total Number of Data Points

**How to Calculate Variance?**

The steps to calculate the variance of a dataset:

**Step1: Calculate the Mean (Average):**

Add up all the values in the dataset and divide by the total number of values. This gives you the mean (μ).

***Mean (μ) = (Sum of All Values) / (Total Number of Values)***

**Step2: Find the Squared Differences from Mean:**

For each value in the dataset, subtract the mean calculated in the first step from that value, and then square the result. This gives you the squared difference for each value.

***Squared Difference for Each Value = (Value – Mean)2***

**Step3: Calculate the Average of the Squared Differences:**

Add up all the squared differences calculated in the previous step, and then divide by the total number of values in the dataset. This gives you the variance (σ^2).

***Variance (σ2) = (Sum of all Squared Differences) / (Total Number of Values)***

**Standard Deviation of Ungrouped Data**

For ungrouped data, the standard deviation can be calculated using three methods that are,

- *Actual Mean Method*
- *Assumed Mean Method*
- *Step Deviation Method*

**Standard Deviation by Actual Mean Method**

**Standard Deviation by actual mean method uses the basic mean formula to calculate the mean of the given data** and using this mean value we find out the standard deviation of the given data values. **We calculate the mean in this method with the formula,**

***μ = (Sum of Observations)/(Number of Observations)***

and **then the standard deviation is calculated using the standard deviation formula.**

$$\sigma = \sqrt{\sum_{i}^{n} (x_i - \bar{x})^2/n}$$

**Example: Find Standard Deviation of data set. X = {2, 3, 4, 5, 6}**

*Given,*

- $n = 5$
- $x_i = \{2, 3, 4, 5, 6\}$

*We know,*

*Mean(μ) = (Sum of Observations)/(Number of Observations)*

$\Rightarrow \mu = (2 + 3 + 4 + 5 + 6)/5$

$\Rightarrow \mu = 4$

$\sigma^2 = \sum_{i}^{n} (x_i - \bar{x})^2/n$

$\Rightarrow \sigma^2 = 1/n[(2 - 4)^2 + (3 - 4)^2 + (4 - 4)^2 + (5 - 4)^2 + (6 - 4)^2]$

$\Rightarrow \sigma^2 = 10/5 = 2$

***Thus,*** $\sigma = \sqrt{(2)} = 1.414$

**Standard Deviation by Assumed Mean Method**

For very large values of x finding the mean of the grouped data is a tedious task so we assumed an arbitrary value (A) as the mean value and then calculated the standard deviation using the normal method. Suppose for the group of n data values ( $x_1, x_2, x_3, \ldots, x_n$), the assumed mean is A then the deviation is,

$$d_i = x_i - A$$

Now, assumed mean formula is,

$$\sigma = \sqrt{\sum_{i}^{n} (d_i)^2/n}$$

**Standard Deviation by Step Deviation Method**

We can also calculate the standard deviation of the grouped data using the step deviation method. As in the above method in this method also, we also choose some arbitrary data value as the assumed mean (say A). Then we calculate the deviations of all data values ($x_1, x_2, x_3, \ldots, x_n$), $d_i = x_i - A$

In next step, we calculate the Step Deviations (d') using

$$d' = d/i$$

where '**i**' is a Common Factor of all 'd' Values

Then, **standard deviation formula is,**

$$\sigma = \sqrt{[(\sum(d')^2/n) - (\sum d'/n)^2]} \times i$$

where '**n**' is Total Number of Data Values

**Standard Deviation of Discrete Grouped Data**

In grouped data first, we made a frequency table and then any further calculation was made. For discrete grouped data, the standard deviation can also be calculated using three methods that are,

- Actual Mean Method
- Assumed Mean Method
- Step Deviation Method

Standard Deviation Formula Based on Discrete Frequency Distribution

For a given data set if it has n values ($x_1, x_2, x_3, \ldots, x_n$) and the frequency corresponding to them is ($f_1, f_2, f_3, \ldots, f_n$) then its standard deviation is calculated using the formula,

$$\sigma = \sqrt{\sum_{i}^{n} f_i(x_i - \bar{x})^2/n}$$

where,

- **n** is Total Frequency ($n = f_1 + f_2 + f_3 + \ldots + f_n$ )
- $\bar{x}$ is Mean of Data

**Example: Calculate the standard deviation for the given data**

| xi | fi |
|---|---|
| 10 | 1 |

| xi | fi |
|---|---|
| 4 | 3 |
| 6 | 5 |
| 8 | 1 |

**Solution:**

*Mean ($\bar{x}$) = ∑(fi xi)/∑(fi)*

*⇒ Mean (μ) = (10×1 + 4×3 + 6×5 + 8×1)/(1+3+5+1)*

*⇒ Mean (μ) = 60/10 = 6*

*n = ∑(fi) = 1+3+5+1 = 10*

| xi | fi | fixi | (xi – x̄) | (xi – x̄)2 | fi(xi – x̄)2 |
|---|---|---|---|---|---|
| 10 | 1 | 10 | 4 | 16 | 16 |
| 4 | 3 | 12 | -2 | 4 | 12 |
| 6 | 5 | 30 | 0 | 0 | 0 |
| 8 | 1 | 8 | 2 | 4 | 8 |

*Now,*

***σ = √(∑in fi(xi – x̄)2/n)***

*⇒ σ = √[(16 + 12 + 0 +8)/10]*

*⇒ σ = √(3.6) = 1.897*

*Standard Derivation(σ) = 1.897*

Standard Deviation of Discrete Data by Assumed Mean Method

In grouped data, if the values in the given data set are very large, then we assumed a random value (say A) as the mean of the data. Then the deviation of each value from the assumed mean is calculated as,

***di = xi – A***

Now formula for standard deviation by assumed mean method is,

***σ = √[(∑(fidi)2 /n) – (∑fidi/n)2]***

where,

- ‘**f**‘ is Frequency of Data Value x
- ‘**n**‘ is Total Frequency **[n = ∑(fi)]**

Standard Deviation of Discrete Data by Step Deviation Method

We can also use the step deviation method to calculate the standard deviation of the discrete grouped data. As in the above method in this method also, we also choose some arbitrary data value as the assumed mean (say A). Then we calculate the deviations of all data values (x1, x2, x3, …, xn), di = xi – A

In next step, **we calculate the Step Deviations (d') using**

***d' = d/i***

where '**i**' is Common Factor of all '**d**' values

Then, **standard deviation formula is,**

$$\sigma = \sqrt{[(\sum(fd')2 /n) - (�'/n)2]} \times i$$

where '**n**' is Total Number of Data Values

**Standard Deviation of Continuous Grouped Data**

For the continuous grouped data we can easily calculate the standard deviation using the Discrete data formulas by replacing each class with its midpoint (as xi) and then normally calculating the formulas.

**Midpoint of each class is calculated using formula,**

***xi (Midpoint) = (Upper Bound + Lower Bound)/2***

**For example, Calculate standard deviation of continuous grouped data as given in table,**

| **Class** | 0-10 | 10-20 | 20-30 | 30-40 |
|---|---|---|---|---|
| **Frequency(fi)** | 2 | 4 | 2 | 2 |

**Solution:**

| **Class** | 5-15 | 15-25 | 25-35 | 35-45 |
|---|---|---|---|---|
| **xi** | 10 | 20 | 30 | 40 |
| **Frequency(fi)** | 2 | 4 | 2 | 2 |

*Mean* $(\bar{x}) = \sum(fi\ xi)/\sum(fi)$

$\Rightarrow$ *Mean* $(\mu) = (10\times2 + 20\times4 + 30\times2 + 40\times2)/(2+4+2+2)$

$\Rightarrow$ *Mean* $(\mu) = 240/10 = 24$

$n = \sum(fi) = 2+4+2+2 = 10$

| **xi** | **fi** | **fixi** | **(xi – x̄)** | **(xi – x̄)2** | **fi(xi – x̄)2** |
|---|---|---|---|---|---|
| 10 | 2 | 20 | 14 | 196 | 392 |
| 20 | 4 | 80 | -4 | 16 | 64 |
| 30 | 2 | 60 | 6 | 36 | 72 |
| 40 | 2 | 80 | 16 | 256 | 512 |

*Now,*

$\boldsymbol{\sigma = \sqrt{(\sum in\ fi(xi - \bar{x})2/n)}}$

$\Rightarrow \sigma = \sqrt{[(392 + 64 + 72 + 512)/10]}$

*⇒ $\sigma = \sqrt{(104)} = 10.198$*
*Standard Derivation($\sigma$) = 10.198*

**Standard Deviation of Probability Distribution**

In probability of all the possible outcomes is generally equal and we take many trials to find the experimental probability of the given experiment.

- For a normal distribution, the mean expected mean is zero and the Standard Deviation is 1.
- For a binomial distribution, the standard deviation is given by the formula,

$$\sigma = \sqrt{(npq)}$$

*where,*

- ***n*** *is Number of Trials*
- ***p*** *is Probability of Success of Trial*
- ***q*** *is Probability of Failure of Trial ($q = 1 - p$)*
- For a Poisson distribution, standard deviation is given by

$$\sigma = \sqrt{\lambda t}$$

*where,*

- *$\lambda$ is Average Number of Successes*
- ***t*** *is Given Time Interval*

**Standard Deviation of Random Variables**

Random variables are the numerical values that denote the possible outcome of the random experiment in the sample space. **Calculating the standard deviation of the random variable tells us about the probability distribution of the random variable and the degree of the difference from the expected value.**

We use **X, Y, and Z as function to represent the random variables.** The probability of the random variable is denoted as P(X) and the expected value is denoted by the μ symbol.

**Then standard deviation of probability distribution is given using formula,**

$$\sigma = \sqrt{(\sum (x_i - \mu)2 \times P(X)/n)}$$

Articles related to Standard Deviation:

- *Mean*
- *Mode*
- *Mean Deviation*

**Standard Deviation Formula Example**

**Example 1: Find the Standard Deviation of the following data,**

| | | | |
|---|---|---|---|
| **xi** | 5 | 12 | 15 |
| **fi** | 2 | 4 | 3 |

**Solution:**

First make the table as follows, so we can calculate the further values easily.

| Xi | fi | Xi×fi | Xi-μ | (Xi-μ)2 | f×(Xi-μ)2 |
|---|---|---|---|---|---|
| 5 | 2 | 10 | -6.375 | 40.64 | 81.28 |
| 12 | 3 | 36 | 0.625 | 0.39 | 1.17 |
| 15 | 3 | 45 | 3.625 | 13.14 | 39.42 |
| **Total** | 8 | 91 | | | 121.87 |

*Mean ($\mu$) = $\sum(f_i x_i)/\sum(f_i)$*
*⇒ Mean ($\mu$) = 91/8 = 11.375*

*$\sigma = \sqrt{(\sum in\ fi(xi - \mu)2/n)}$*
*$\Rightarrow \sigma = \sqrt{[(121.87)/(8)]}$*
*$\Rightarrow \sigma = \sqrt{(15.234)}$*
*$\Rightarrow \sigma = 3.90$*
*Standard Derivation(σ) = 3.90*

**Example 2: Find Standard Deviation of following data table.**

| Class | Frequency |
|---|---|
| 0-10 | 3 |
| 10-20 | 6 |
| 20-30 | 4 |
| 30-40 | 2 |
| 40-50 | 1 |

**Solution:**

| Class | Xi | fi | f×Xi | Xi – μ | (Xi – μ)2 | f×(Xi – μ)2 |
|---|---|---|---|---|---|---|
| 0-10 | 5 | 3 | 15 | -15 | 225 | 675 |
| 10-20 | 15 | 6 | 90 | -5 | 25 | 150 |
| 20-30 | 25 | 4 | 100 | 5 | 25 | 100 |
| 30-40 | 35 | 2 | 70 | 15 | 225 | 450 |
| 40-50 | 45 | 1 | 45 | 25 | 625 | 625 |
| **Total** | | **16** | **320** | | | **2000** |

*Mean $(\mu) = \sum(fi\ xi)/\sum(fi)$*
*$\Rightarrow$ Mean $(\mu) = 320/16 = 20$*
*$\sigma = \sqrt{(\sum in\ fi(xi - \mu)2/n)}$*
*$\Rightarrow \sigma = \sqrt{[(2000)/(16)]}$*
*$\Rightarrow \sigma = \sqrt{(125)}$*
*$\Rightarrow \sigma = 11.18$*
*Standard Derivation(σ) = 11.18*

For a comprehensive collection of maths formulas across different grade levels and concepts, keep following GeeksforGeeks.

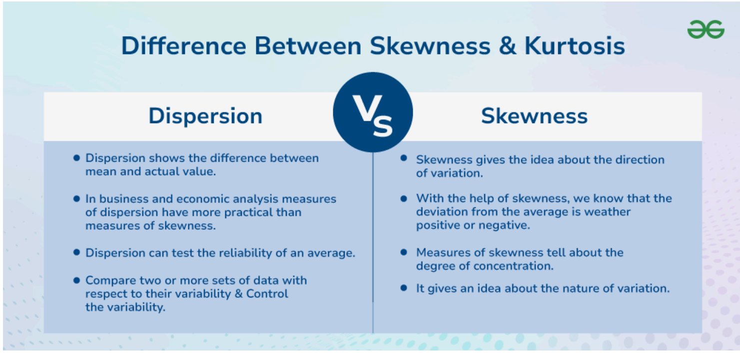
**Standard Deviation Formula in Excel**

- Use Excel's built-in functions STDEV.P for the entire population or STDEV.S for a sample.

- **Step-by-Step Guide:** Enter your data set in a single column, then type =STDEV.S(A1:A10) (replace A1:A10 with your data range) in a new cell to get the standard deviation for a sample.
- **Visual Aids:** Utilize Excel's chart tools to visually represent data variability alongside standard deviation.

**What is Skewness?**

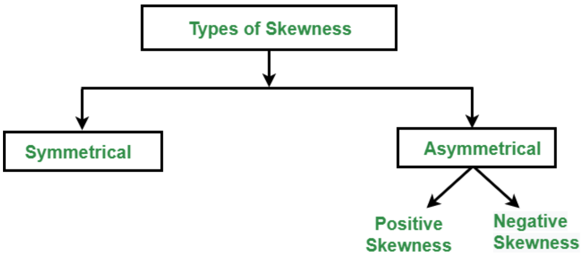
**Skewness** is an important statistical technique that helps to determine the asymmetrical behavior of the frequency distribution, or more precisely, the lack of symmetry of tails both left and right of the frequency curve. A distribution or dataset is symmetric if it looks the same to the left and right of the center point.




*Difference between Skewness and Kurtosis*

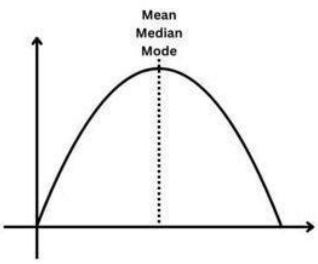
**Types of Skewness**

The following figure describes the classification of skewness:




*Types of Skewness*

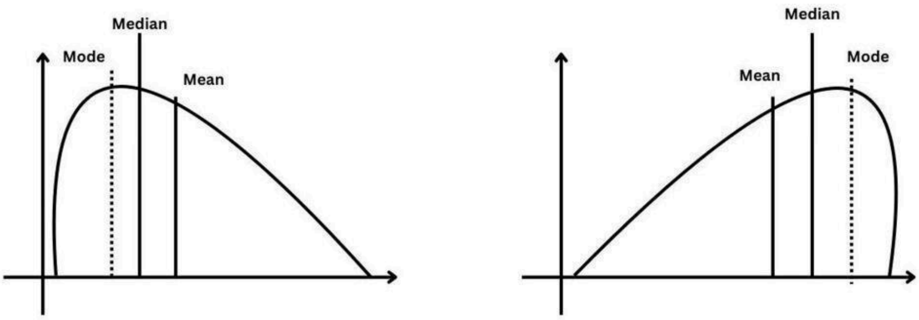
**1. Symmetric Skewness:** A perfect symmetric distribution is one in which frequency distribution is the same on the sides of the center point of the frequency curve. In this, Mean = Median = Mode. There is no skewness in a perfectly symmetrical distribution.



*Symmetric Skewness*

**2. Asymmetric Skewness:** A asymmetrical or skewed distribution is one in which the spread of the frequencies is different on both the sides of the center point or the frequency curve is more stretched towards one side or value of Mean. Median and Mode falls at different points.

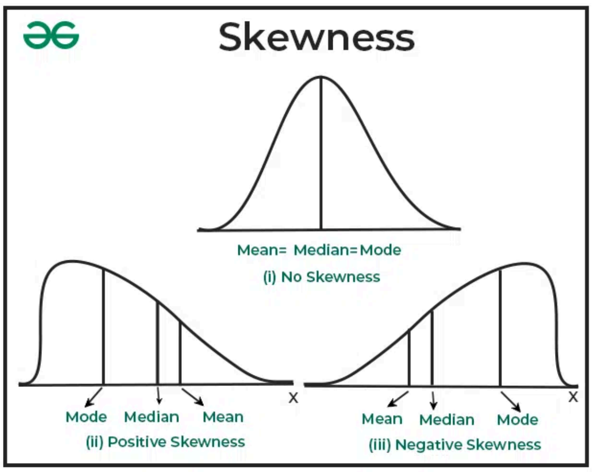
- **Positive Skewness:** In this, the concentration of frequencies is more towards higher values of the variable i.e. the right tail is longer than the left tail.
- **Negative Skewness:** In this, the concentration of frequencies is more towards the lower values of the variable i.e. the left tail is longer than the right tail.
- **Positive Skewness:** In this, the concentration of frequencies is more towards higher values of the variable i.e. the right tail is longer than the left tail.
- **Negative Skewness:** In this, the concentration of frequencies is more towards the lower values of the variable i.e. the left tail is longer than the right tail.




*Positive Skewness and Negative Skewness*

**What is Skewness?**

Skewness can be defined as a statistical measure that describes the lack of symmetry or asymmetry in the probability distribution of a dataset. It quantifies the degree to which the data deviates from a perfectly symmetrical distribution, such as a [normal (bell-shaped) distribution](). Skewness is a valuable statistical term because it provides insight into the shape and nature of a dataset's distribution. **For example,** understanding whether a dataset is positively or negatively skewed can be important in various fields, including [finance](), [economics](), and data analysis, as it can impact the interpretation of data and the choice of statistical techniques.



**Tests of Skewness**
There are several statistical tests and methods to assess the skewness of a dataset. These tests can help you determine whether a dataset is positively skewed, negatively skewed, or approximately symmetric. Here are some common tests and techniques used to assess skewness:
**1. Visual Inspection:** The simplest way to assess skewness is by creating a histogram or a density plot of the given data. If the plot is skewed to the left, it is negatively skewed, and if the plot is skewed to the right, it is positively skewed. If the plot is roughly symmetric, it has no skewness.
**2. Skewness Coefficient (Pearson's First Coefficient of Skewness):** This is a numerical measure of skewness, which determines the skewness when mean and mode are not equal. It is calculated as:
Skewness as per Karl Pearson's Measure

$$Skewness = Mean - Mode$$

*Skewness of Karl Pearson's Measure*

- If mean is greater than mode, the skewness will consist positive value.
- In case of mean is smaller than mode, the skewness will be a negative value.
- In case of equality of mean and mode, the skewness will be zero.

**3. Quartiles are not equidistant** from each other; i.e., Q3−Me≠Me−Q1$Q3-Me$偃$=Me-Q1$

**Positive and Negative Skewness**
Positive skewness and negative skewness are two different ways that a dataset's distribution can deviate from perfect symmetry (a normal distribution). They describe the direction of the skew or asymmetry in the data.
1. Positive Skewness (Right Skew)
In a positively skewed distribution, the tail on the right side (the larger values) is longer than the tail on the left side (the smaller values). This means that the majority of data points are concentrated on the left side of the distribution, and there are some extreme values on the right side. In the case of a positively skewed dataset,

$$Mean > Median > Mode$$

**Examples** of positively skewed data include income distribution (where most people earn a moderate income, but a few earn extremely high incomes), exam scores (where most students score in a certain range, but a few score exceptionally high), and stock market returns (where most days have modest returns, but a few days may have very high returns).
2. Negative Skewness (Left Skew)
In a negatively skewed distribution, the tail on the left side (the smaller values) is longer than the tail on the right side (the larger values). This implies that most of the data points are concentrated on the right side of the distribution, with a few extreme values on the left side. In the case of a negatively skewed dataset,

$$Mean < Median < Mode$$

**Examples** of negatively skewed data include test scores on an easy test (where most students score well, but a few score very low), the age at retirement (where most people retire at a certain age, but a few retire exceptionally early), and the gestational age at birth (where most babies are born full-term, but a few are born prematurely).
3. Zero Skewness (Symmetrical Distribution)
Zero skewness indicates a perfectly symmetrical distribution, where the mean, median, and mode are equal. In a symmetrical distribution, the data points are evenly distributed around the central point.
**Example:** A perfectly balanced dataset with equal frequencies of all values.

**Measurement of Skewness**

**1. Karl Pearson's Measure**
Karl Pearson's Measure of Skewness **uses the mean, median, and standard deviation of the given data set to quantify the asymmetry or lack of symmetry in the distribution**. It is a dimensionless number that provides valuable insights into the shape of a dataset's distribution. This measure is valuable in various fields of statistics and data analysis, helping researchers and analysts understand the direction and degree of skewness in their datasets, which can inform subsequent modeling and analytical decisions.
Skewness as per Karl Pearson's Measure

$$Skewness = Mean - Mode$$

*Skewness of Karl Pearson's Measure*

- If mean is greater than mode, the skewness will consist positive value.
- In case mean is smaller than mode, the skewness will be a negative value.
- In case of equality of mean and mode, the skewness will be zero.

Coefficient of Skewness as per Karl Pearson's Measure

***1. With respect to Mean and Median:*** *Sk=3×(X̄–M)σSk=σ3×(X̄–M)*

***2. With respect to Mean and Mode:*** *Sk=(X̄–Mode)σSk=σ(X̄–Mode)*

*Here,*

- *Sk is Karl Pearson's skewness coefficient.*
- *X̄ X̄ is the arithmetic mean or average of the data.*
- *M is the middle value of the data when it is arranged in ascending order.*
- *σ is a measure of the standard deviation of the data.*

Coefficient of Karl Pearson's Measure

- *If Sk = 0, it indicates a perfectly symmetric distribution where the data is evenly balanced on both sides of the mean.*
- *If Sk > 0, it suggests a positively skewed distribution where the tail on the right side is longer or fatter, and the majority of data points are concentrated on the left side of the mean.*
- *If Sk < 0, it indicates a negatively skewed distribution where the tail on the left side is longer or fatter, and the majority of data points are concentrated on the right side of the mean.*

Example of Karl Pearson's Measure

Calculate Pearson's skewness coefficient for a dataset of exam scores: 85, 88, 92, 94, 96, 98, 100, 100, 100, 100.

*Solution:*

**Step 1:** Calculation of Mean

Mean(X̄)=85+88+92+94+96+98+100+100+100+10010=95310=95.3*Mean*(*X̄*)=1085+88+92+94+96+98+100+100+100+100=10953=95.3

**Mean = 95.3**

**Step 2:** Calculation of Median

Since there are 10 data points, the median is the average of the 5th and 6th values when sorted in ascending order:

Median=(96+98)2=1942=97*Median*=2(96+98)=2194=97

**Median = 97**

**Step 3:** Calculation of standard deviation.

σ2=Σ(xi−μ)2N=(85−95.3)2+…+(100−95.3)210=268.110=26.81*σ*2=*N*Σ(*xi*−*μ*)2=10(85−95.3)2+…+(100−95.3)2=10268.1=26.81

Thus, σ=√26.81

**σ = ~5.**

**Step 4:** Calculation of mode

It is clear from the data set that 100 is the most frequently occurring value in the data. Hence, mode of given data is 100.

**Step 5:** Substitute the values in the formulae

*A. With respect to Mean and Median*

Sk=3(X̄−M)σ=(3×(95.3−97))5=−5.15*Sk*=*σ*3(*X̄*−*M*)=5(3×(95.3−97))=5−5.1

**Sk = -1.02**

*B. With respect to Mean and Mode*

Sk=(X̄−Mode)σ=(95.3−100)5*Sk*=*σ*(*X̄*−*Mode*)=5(95.3−100)

**Sk = -0.94**

**Interpretation:** The skewness coefficient (Sk) is negative, indicating a slight negative skewness in the distribution of exam scores. This means that the tail of the distribution is slightly longer on the left side, and most of the scores are concentrated on the right side of the mean.

**2. Bowley's Measure**

Bowley's Skewness Coefficient, named after the British economist **Arthur Lyon Bowley**, is a **statistical measure used to assess the skewness or asymmetry in a probability distribution**. Unlike some other skewness measures that rely on moments or deviations from the mean, Bowley's

Skewness Coefficient is based on quartiles. This coefficient provides a simple and intuitive way to understand the direction and magnitude of skewness in a dataset. Bowley's Skewness Coefficient is especially useful when dealing with data that may not follow a normal distribution or when a robust measure of skewness is required.

*B = Q1+Q3−2Q2Q3−Q1Q3−Q1Q1+Q3−2Q2*

- *Q1 is the first quartile (25th percentile),*
- *Q2 is the second quartile (50th percentile, or median), and*
- *Q3 is the third quartile (75th percentile).*

Coefficient of Bowley's Measure

- *If B = 0, the distribution is perfectly symmetric about the mean (no skewness).*
- *If B < 0, the distribution is negatively skewed (left-skewed), meaning the tail on the left side of the distribution is longer or heavier.*
- *If B > 0, the distribution is positively skewed (right-skewed), indicating that the tail on the right side of the distribution is longer or heavier.*

Example of Bowley's Measure:

Calculate Bowley's Measure of Skewness for the following dataset representing the ages of a group of people in a sample: 20, 24, 28, 32, 35, 40, 42, 45, 50.

*Solution:*

**Step 1:** Calculate the median (Q2)

**Q2= 35** (the middle value)

**Step 2:** Calculate the first quartile (Q1)

To find Q1, consider the values to the left of the median: 20, 24, 28, 32

Q1=24+282=26*Q*1=224+28=26

**Q1 = 26**

**Step 3:** Calculate the third quartile (Q3)

To find Q3, consider the values to the right of the median: 40, 42, 45, 50.

Q3=42+452=43.5*Q*3=242+45=43.5

**Q3 = 43.5**

**Step 4:** Substitute the above values in the formula

B=Q1+Q3−2Q2Q3−Q1=26+43.5−2×3543.5−26*B*=*Q*3−*Q*1*Q*1+*Q*3−2*Q*2=43.5−2626+43.5−2×35

**B = -0.02**

**Interpretation:** Since B is negative (B < 0), the distribution is negatively skewed (left-skewed). This means that the tail of the distribution is longer on the left side, indicating that there may be outliers or high values on the right side of the data.

**3. Kelly's Measure**

Kelly's measure of skewness is **a way to quantify the degree of skewness in a distribution by comparing the values of certain percentiles (typically the 10th, 50th, and 90th percentiles) or deciles (10th, 20th, …, 90th percentiles) of the dataset**. Specifically, it involves comparing the difference between the median (50th percentile) and the average of the 10th and 90th percentiles (or deciles) to assess the skewness of the data.

Skewness as per Kelly's Measure

*Skewness=P90+P10−2P502Skewness=2P90+P10−2P50*

Coefficient of Skewness as per Kelly's Measure

*SKL = 90thpercentile+10thpercentile−2×50thpercentile90thpercentile−10thpercentile90thpercentile−10thpercentile90thpercentile+10thpercentile−2×50thpercentile*

Coefficient of Kelly's Measure

- *If SKL is positive, it indicates positive skewness, meaning the distribution has a longer right tail.*
- *If SKL is negative, it indicates negative skewness, meaning the distribution has a longer left tail.*
- *If SKL is close to zero, it suggests that the distribution is approximately symmetric.*

Example of Kelly's Measure:

Calculate Kelly's Coefficient of Skewness for the following data: 5, 7, 8, 9, 10, 12, 15, 16, 18, 20.

*Solution:*

**Step 1:** Find the 10th Percentile

To find the 10th percentile, we need to rank the data in ascending order and find the value below which 10% of the data falls. In this dataset, the 10th percentile corresponds to the value at position 1 since 10% of 10 data points is 1. So, the 10th percentile is **5**.

**P10 = 5**

**Step 2:** Find the 50th Percentile (Median)

Since there are 10 data points, the median is the average of the 5th and 6th values when sorted in ascending order

Median=10+122=222=11*Median*=210+12=222=11

**P50 = 11**

**Step 3:** Find the 90th Percentile

To find the 90th percentile, you need to identify the value below which 90% of the data falls. In this dataset, the 90th percentile corresponds to the value at position 9 since 90% of 10 data points is 9. So, the 90th percentile is 18.

**P90 = 18**

**Step 4:** Substitute the values in the formula.

SKL=18+5−2×1118−5*SKL*=18−518+5−2×11

**SKL = 0.07**

**Interpretation:** Since the value is positive, Kelly’s Skewness Coefficient suggests that the distribution has a slight positive skewness, meaning it has a longer right tail. This indicates that there may be some data points on the right side of the distribution that are relatively larger compared to the majority of data points.

**Interpretation of Skewness**

Interpreting skewness involves understanding the nature of the skewness (left or right) and its magnitude. Here is how to interpret skewness:

Direction of Skewness

**Negative Skewness (Left Skewed):** If the skewness is negative, it indicates that the distribution is skewed to the left. In a left-skewed distribution:

- The tail on the left side (the smaller values) is longer and often contains outliers.
- The majority of data points are concentrated on the right side.
- The mean is typically less than the median.

**Positive Skewness (Right Skewed):** A positive skewness indicates that the distribution is skewed to the right. In a right-skewed distribution:

- The tail on the right side (the larger values) is longer and may contain outliers.
- Most data points are concentrated on the left side.
- The mean is typically greater than the median.

**Zero Skewness (Symmetric):** A skewness value close to zero suggests a symmetric distribution where the data is evenly distributed on both sides of the mean. This means there is no skewness.

Magnitude of Skewness

The magnitude of skewness provides information about the degree of skewness.

- If the skewness value is close to 0 (between -0.5 and 0.5), the distribution is approximately symmetric.
- If the skewness value is significantly negative (below -1), it suggests strong left skewness.
- If the skewness value is significantly positive (above 1), it suggests strong right skewness.
    - **Difference between Dispersion and Skewness**

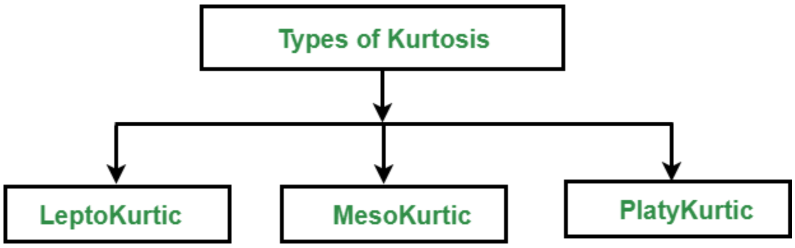
| Basis | Dispersion | Skewness |
|---|---|---|
| **Focus** | Focuses on the variability or spread of data points around the central tendency (mean, median). | Focuses on the shape of the distribution and the direction of the skewness (left or right). |

| Basis | Dispersion | Skewness |
|---|---|---|
| **Measurement** | Common measures of dispersion include variance, standard deviation, range, and interquartile range (IQR). | Common measures of skewness include Pearson's first coefficient of skewness, moment skewness, and graphical methods like Q-Q plots. |
| **Relationship to Mean** | Dispersion measures are not directly related to the mean, although they can affect the mean's interpretation when it is used as a measure of central tendency. | Skewness provides information about the relationship between the mean and the median. Positive skewness implies that the mean is greater than the median, and negative skewness implies the opposite. |
| **Interpretation** | High dispersion indicates that data points are scattered or dispersed widely from the center, suggesting a wide range of values. | Positive skewness indicates a right-skewed distribution with a longer right tail, while negative skewness indicates a left-skewed distribution with a longer left tail. Zero skewness suggests a symmetric distribution. |
| **Application** | Dispersion measures are useful for understanding the variability of data and assessing how tightly or loosely data points are clustered around the central value. | Skewness is useful for understanding the shape of a distribution and identifying whether it is skewed to the left or right. It helps assess the asymmetry of the data. |
| **Examples** | Examples of dispersion include the spread of test scores in a classroom, the variability of stock prices over time, or the range of ages in a population. | Examples of skewness include income distributions (often right-skewed), response times for a website (potentially right-skewed with outliers), and exam score distributions (which can be either left-skewed or right-skewed). |

- **Summary**
- Skewness is a key statistical measure that describes the asymmetry of a distribution. It can be positive, negative, or zero, indicating the direction and degree of skewness. Measures of skewness, such as Pearson's coefficients and the moment coefficient, help quantify this asymmetry. Interpreting skewness is essential for understanding the distribution of data and making informed decisions.

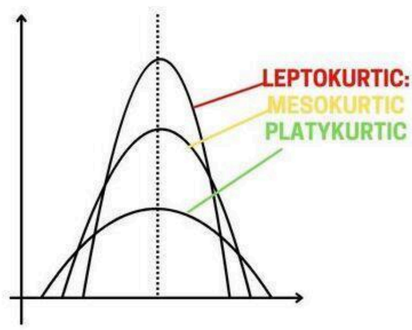
**What is Kurtosis?**
It is also a characteristic of the frequency distribution. It gives an idea about the **shape of a frequency distribution**. Basically, the measure of kurtosis is the extent to which a frequency distribution is peaked in comparison with a normal curve. It is the degree of peaked Ness of a distribution.

**Types of Kurtosis**
The following figure describes the classification of kurtosis:




*Types of Kurtosis*

1. **Leptokurtic:** Leptokurtic is a curve having a high peak than the normal distribution. In this curve, there is too much concentration of items near the central value.
2. **Mesokurtic:** Mesokurtic is a curve having a normal peak than the normal curve. In this curve, there is equal distribution of items around the central value.
3. **Platykurtic:** Platykurtic is a curve having a low peak than the normal curve is called platykurtic. In this curve, there is less concentration of items around the central value.




*Types of Kurtosis*

**Difference Between Skewness and Kurtosis**
Below is the difference between Skewness and Kurtosis.

| Sr. No. | Skewness | Kurtosis |
|---|---|---|
| 1. | It indicates the shape and size of variation on either side of the central value. | It indicates the frequencies of distribution at the central value. |
| 2. | The measure differences of skewness tell us about the magnitude and direction of the asymmetry of a distribution. | It indicates the concentration of items at the central part of a distribution. |

| Sr. No. | Skewness | Kurtosis |
|---|---|---|
| 3. | It indicates how far the distribution differs from the normal distribution. | It studies the divergence of the given distribution from the normal distribution. |
| 4. | The measure of skewness studies the extent to which deviation clusters is are above or below the average. | It indicates the concentration of items. |
| 5. | In an asymmetrical distribution, the deviation below or above an average is not equal. | No such distribution takes place. |

**Conclusion**

Understanding skewness and kurtosis is crucial for analyzing and interpreting data distributions. Skewness helps identify the asymmetry of a distribution, indicating whether data points are more concentrated on one side of the mean. On the other hand, kurtosis measures the "peakedness" or "tailedness" of a distribution. Both skewness and kurtosis provide valuable insights into the shape and characteristics of data, enabling more accurate data analysis and decision-making.

**Skewness Formula**

The measure of skewness tells us the direction and the extent of skewness. In symmetrical distribution the mean, median, and mode are identical. the more the mean moves away from the mode, the larger the asymmetric or skewness. Before learning let's learn more about Mean, Median, and Mode first.

**Mean**

Mean is the average of the numbers in the data distribution, It is calculated by adding up all the values in the dataset and dividing the sum by the number of values in the dataset.

***Mean= Sum of all values in Dataset / Total number of values***

**Example: Find the mean of a dataset of exam scores: 70, 80, 85, 90, and 95.**

**Solution:**

*Mean = (70 + 80 + 85 + 90 + 95) / 5 = 84*

*So the mean of this dataset is 84.*

**Median**

When arranging all the data in order (ascending and descending) the comes in the middle of the data is called the median.

**Median is the middle value of a dataset when the values are arranged in order from smallest to largest.**

Examples for Odd Numbers in the Dataset

**Example 1: Find the median of a dataset of exam scores: 70, 85, 80, 95, 90**

**Solution:**

*Firstly arrange all no. in order from smallest to largest: 70, 80, 85, 90, 95.*

*The mid value is 85. so, the median is 85.*

**Example 2: Find the median of a dataset: 5, 10, 15, 20, 25.**

**Solution:**

*Firstly arrange all no. in order from smallest to largest: 5, 10, 15, 20, 25.*

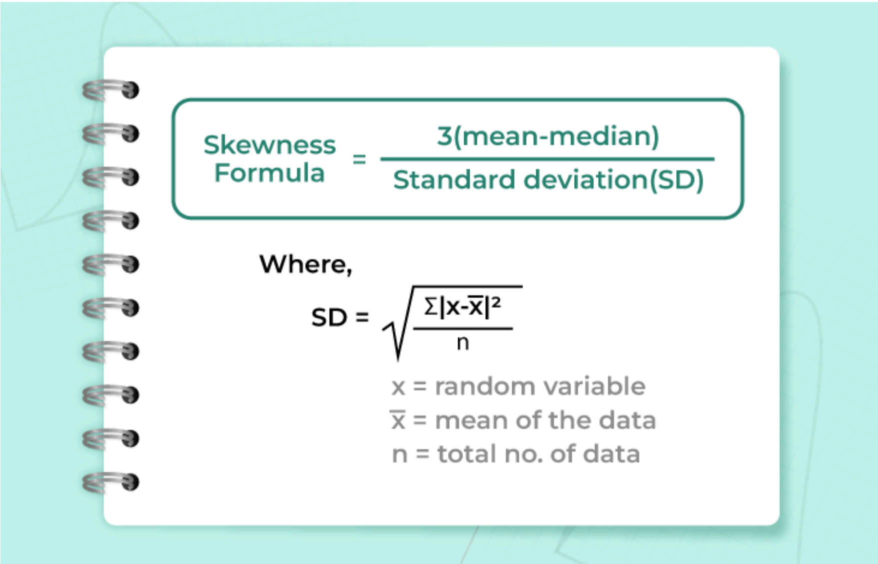
*The mid value is 15. so, the median is 15.*

**If there are an even number of values in the dataset, the median is calculated by taking the average of the two middle values.**

Examples for Even Numbers in the Dataset

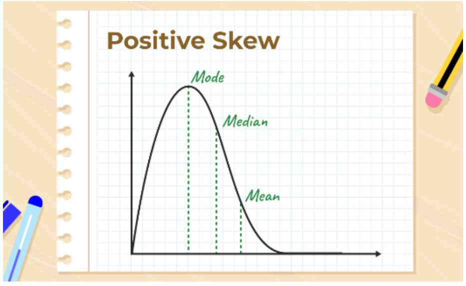
**Example 1: Find the median of a dataset of exam scores: 70, 80, 85, 90.**

**Solution:**
*The median is calculated as (80 + 85) / 2 = 82.5*
*So the median of this dataset is 82.5.*
**Example 2: Find the median of a dataset: 2, 4, 6, 8, 10, 12.**
**Solution:**
*Firstly, we need to find the middle two numbers. So, 6, and 8 are mid values of the dataset*
*Median = (6 + 8) / 2 = 7*
*So the median of this dataset is 7.*
**Mode**
most frequently used number in data is called the mode of the data.
**Example 1: We have a data set representing the number of pets owned by 10 people: 3, 1, 0, 2, 1, 1, 4, 2, 2, 1. Find the mode.**
**Solution:**
So, the value that appears most frequently in the data set is 1. the value 1 appears four times.
Therefore, the mode of this data set is 1.
**Skewness Formula**
The skewness formula is discussed in the image below,




**Type of Skewness**
Various types of skewness used in mathematics are,
- Positive Skewness
- Negative Skewness
- Zero Skewness

Positive Skewness
Positive Skewness means the tail on the right side of the distribution is longer. The mean and median will be greater than the mode.
Condition for positive skewness = **Mean > Median >Mode**
The positive curve of skewness is shown in the image below,



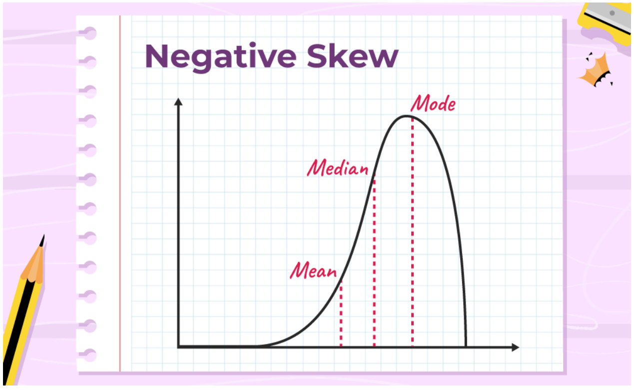

Let’s take an example of the income distribution where a few people earn very high incomes and the majority earn lower incomes. so, this is often positively skewed. Analyzing skewed data can provide valuable insights into the underlying causes and potential solutions or interventions.

Negative Skewness

Negative Skewness means when the tail of the left side of the distribution is longer than the tail on the right side. The mean and median will be less than the mode.

Condition for negative skewness is **Mode > Median > Mean**

The curve shows negative skewness in the image below,



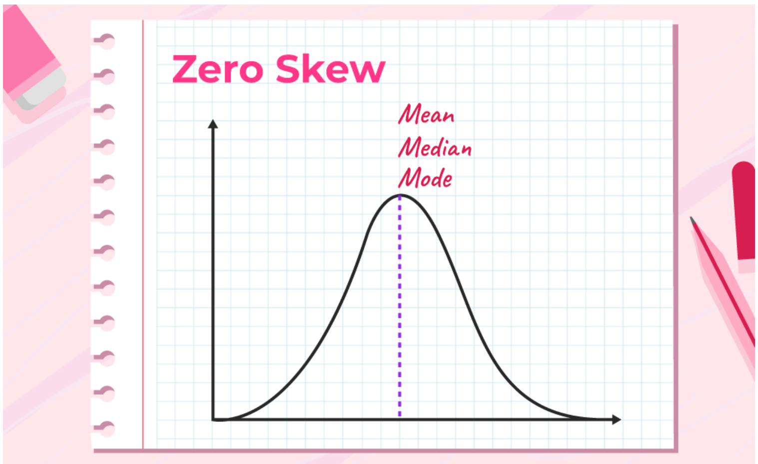

Let’s take an example of a match, during the match most of the players of a particular team scored runs above 50 and only a few of them scored below 10. In such a case, the data is generally represented with the help of a negatively skewed distribution. And this data is helpful to analyze the game’s performance.

Zero Skewness

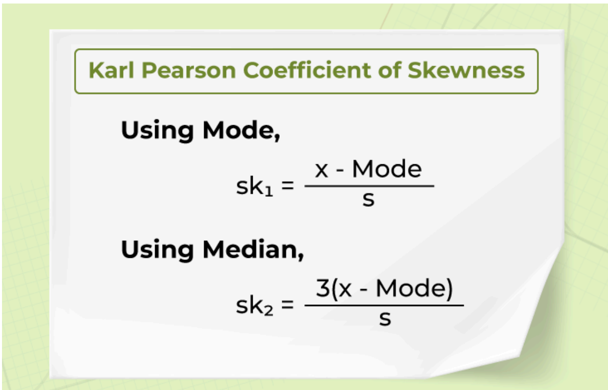
It is also known as a “symmetric distribution”.It signifies that distribution of data is evenly distributed around the mean, with no long tails on either end of the distribution

Condition for zero skewness is **Mean = Mode = Median**

The curve for zero skews is shown in the image below,




**Methods to Measure Skewness**
Skewness can be measured using Karl Pearson's Coefficient of Skewness.
Karl Pearson's Co-efficient of Skewness
The formula for measuring Skewness using Karl Pearson's Co-efficient is discussed below in the image,




**Conditions**
- Mean = Mode = Median, then the coefficient of skewness is zero for symmetrical distribution.
- Mean > Mode, then the coefficient of skewness will be positive.
- Mean < Mode, then the coefficient of skewness will be negative.

Karl person`s coefficient of skewness has a positive sign for the positively skewed and a negative sign for the negatively skewed.
**Read More**,
- *Statistics*
- *Coefficient of Skewness*

**Solved Examples on Skewness Formula**
**Example 1: Find the skewness for the given Data ( 2,4,6,6)**
**Solution:**
*Mean of Data = (2 + 4 + 6 + 6) / 4*
*= 18 / 4*
*= 4.5*
*Number of terms (n) = 4 (even)*
*Median of Data = {[n / 2]th + [n / 2 + 1]th}/2 term*
*= [(4 /2)th term + (4/2 +1)th term] / 2*
*= [2nd term + 3rd term] / 2*
*= [4+6]/2*

*= 10/2*
*Median of Data = 5*
*Mode of Data = Highest Frequency term = 6 (frequency 2)*
*S.D. = √[(2 – 5)2 + (4-5)2 + (6-5)2 + (6-5)2/4]*
*= √[(9 + 1 + 1 + 1)/4]*
*= √(3)*
*= 1.732*

***Skewness = 3(Mean – Median)/S.D.***

*By Applying Skewness Formula,*
*Skewness = 3(4.5 – 5)/1.732*
*= 3(-0.5)/ 1.732*
*Skewness = – 0.866*
*So, the skewness of these data is negative.*

**Example 2: A boy collects some rupees in a week as follows (25,28,26,30,40,50,40) and finds the skewness of the given Data in question with the help of the skewness formula.**

**Solution:**

*Mean of Data = (25+28+26+30+40+50+40) / 7*
*= 239 / 7*
*= 34.14*
*Number of terms (n) =7 (odd)*
*Arrange Data in ascending order = 25,26 ,28,30,40,40,50*
*The median of data is = 30*
*Mode of Data = Highest Frequency term = 40 (frequency 2)*
*S.D = √(1/7 – 1) x ((25 – 34.1429)2 + (28 – 34.1429)2 + (26 – 34.1429)2 + (30 – 34.1429)2 + (40 – 34.1429)2 +(534.1429)2 + (40 – 34.1429)2)*
*= √(1/6) x ((-9.1429)2 + (-6.1429)2 + (-8.1429)2 + (-4.1429)2 + (5.8571)2 + (15.8571)2 + (5.8571)2)*
*= √(0.1667) x ((83.5926) + (37.7352) + (66.3068) + (17.1636) + (34.3056) + (251.4476) + (34.3056))*
*= √(0.1667) x 524.8571*
*= √87.4762*
*. = 9.3529*

***Skewness = 3(Mean – Median)/S.D.***

*By Applying Skewness Formula,*
*Skewness = 3(34.14 – 30)/9.3529*
*= 1.32*
*Skewness = 1.32*
*So skewness for these data is positive*

**Example 3: Attendance of all classes of a school are as follows find their skewness?**
**1st (35), 2nd(32), 3rd(38), 4th(39), 5th(43)**

| Class Name | Number of students |
|---|---|
| 1st | 35 |
| 2nd | 32 |
| 3rd | 38 |

| Class Name | Number of students |
|---|---|
| 4th | 39 |
| 5th | 45 |

**Solution:**

*Mean of Data = (35 + 32 + 38 + 39 + 42)/5*
*= 186/5*
*= 37.2*
*Number of terms (n) = 5 (odd)*
*Arrange Data in ascending order = 32,35,38,39,42*
*Median of Data = 38*
*S.D. = √(1/5 – 1) x ((35 – 37.2)2 + (32 – 37.2)2 + (38 – 37.2)2 + (39 – 37.2)2 + (42 – 37.2)2)*
*= √(1/4) x ((-2.2)2 + (-5.2)2 + (0.8)2 + (1.8)2 + (4.8)2)*
*= √(0.25) x ((4.84) + (27.04) + (0.64) + (3.24) + (23.04))*
*= √(0.25) x 58.8*
*= √14.7*
*= 3.8341*

***Skewness = ∑(yi – ymean) / (n – 1) x (sd)³***

*Skewness =((35 – 37.2)³ + (32 – 37.2)³ + (38 – 37.2)³ + (39 – 37.2)³ + (42 – 37.2)³) / (5 – 1)³ x 3.8341*
*Skewness = ((-2.2)³ + (-5.2)³ + (0.8)³ + (1.8)³ + (4.8)³ )/ (4)³ x 3.8341*
*Skewness =((-10.648) + (-140.608) + (0.512) + (5.832) + (110.592)) / 64 x 3.8341*
*Skewness =-34.32 / 245.3824*
*Skewness = -0.1522*
*So, the skewness of these data is negative.*

**Can skewness be zero?**

*Yes, skewness can be zero. This occurs when the distribution is perfectly symmetrical. It is possible if there are an equal number of values on the left and right sides of the mean.*

What does positive skewness mean?

*Positive skewness means the distribution is skewed to the right. There are more values on the right side of the mean than on the left side.*

***Condition for Positive Skewness = Mean > median >mode***

What does negative skewness mean?

*Negative skewness means the distribution is skewed to the left. There are more values on the left side of the mean than on the right side.*

***Condition for negative skewness = Mode > median > mean***

How is skewness used in finance and investment analysis?

*In finance and investment analysis, skewness is used to measure the degree of asymmetry in returns on investment. Skewed returns can have an impact on portfolio management and risk management strategies, and understanding the skewness of a particular investment can help investors to make better-informed decisions.*

What is the formula for calculating skewness?

*The formula used to calculate Skewness is,*

***Skewness Formula= 3(Mean – Median)/S.D.***

How is skewness used in hypothesis testing?

*Skewness is used in hypothesis testing to determine whether a sample of data is normally distributed or not. If the skewness value is close to zero, the data is likely to be normally distributed. If the* *skewness value is positive or negative, the data is likely to be skewed and may require non-parametric tests for hypothesis testing.*

**Skewness and Kurtosis in R Programming**

In statistics, **skewness** and **kurtosis** are the measures that tell about the shape of the data distribution, or simply, both are numerical methods to analyze the shape of data set unlike, plotting graphs and histograms which are graphical methods. These are normality tests to check the irregularity and asymmetry of the distribution.
To calculate skewness and kurtosis in R language, a **moments** package is required.
Skewness
Skewness is a statistical numerical method to measure the asymmetry of the distribution or data set. It tells about the position of the majority of data values in the distribution around the mean value. A fundamental statistical notion called skewness quantifies the asymmetries in data distributions. It is essential to a number of disciplines, including data analysis, social sciences, economics, and finance.

Formula:
*represents mean of data vector*
***n** represents total number of observations*

There exist 3 types of skewness values on the basis of which the asymmetry of the graph is decided. These are as follows:

Positive Skew
The asymmetry of data distributions where the tail extends towards higher values is known statistically as positive skewness. It is a crucial metric in a number of disciplines, including data analysis, social sciences, finance, and economics. The definition, computation techniques, interpretation, and applications of positive skewness theory are covered in detail in this article.

If the coefficient of skewness is greater than 0 i.e. **\gamma_{1}>0** , then the graph is said to be positively skewed with the majority of data values less than the mean. Most of the values are concentrated on the left side of the graph.
**Example:**
**R**

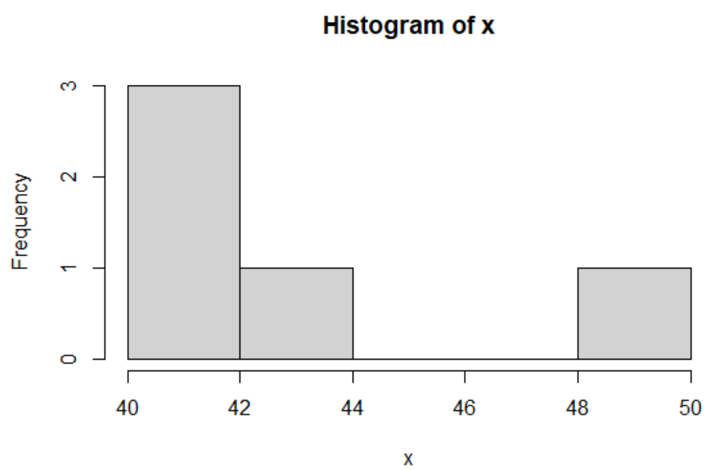
```
# Required for skewness() function

library(moments)



# Defining data vector
```

```
x <- c(40, 41, 42, 43, 50)

# output to be present as PNG file
png(file = "positiveskew.png")

# Print skewness of distribution
print(skewness(x))

# Histogram of distribution
hist(x)

# Saving the file
dev.off()
```

**Output:**

```
[1] 1.2099
```

**Graphical Representation:**



*Simple Histogram*

Zero Skewness or Symmetric
A statistical notion called zero skewness, commonly referred to as symmetry, defines data distributions that are balanced and have equal probability on both sides of the mean. It is a fundamental metric used in many disciplines, such as data analysis, economics, social sciences, and finance.

If the coefficient of skewness is equal to 0 or approximately close to 0 i.e. , then the graph is said to be symmetric and data is normally distributed.
**Example:**
**R**

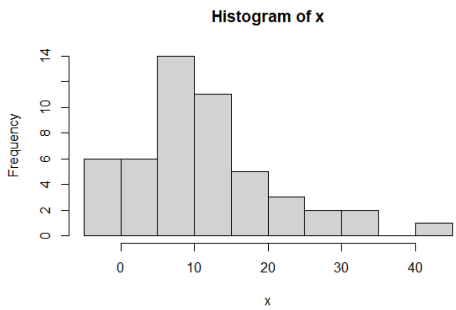
```
# Required for skewness() function

library(moments)



# Defining normally distributed data vector

x <- rnorm(50, 10, 10)



# output to be present as PNG file

png(file = "zeroskewness.png")



# Print skewness of distribution

print(skewness(x))



# Histogram of distribution

hist(x)
```

```
# Saving the file

dev.off()
```

**Output:**

```
[1] -0.02991511
```

**Graphical Representation:**




*Simple Histogram*

Negatively skewed

Left-skewed distributions, commonly referred to as negatively skewed distributions, are statistical notions that describe asymmetrical data distributions with a tail that slopes downward. In a number of disciplines, including finance, economics, social sciences, and data analysis, it is crucial to comprehend negatively skewed data.

If the coefficient of skewness is less than 0 i.e. **\gamma_{1}<0** , then the graph is said to be negatively skewed with the majority of data values greater than the mean. Most of the values are concentrated on the right side of the graph.

**Example:**

**R**

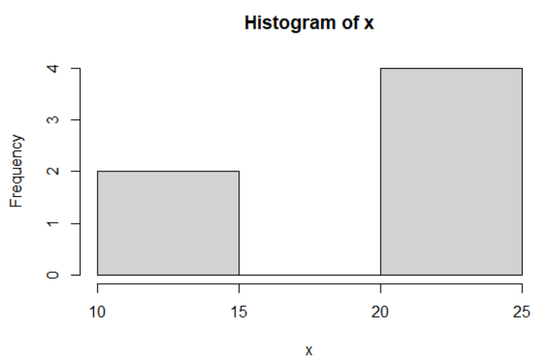
```
# Required for skewness() function

library(moments)



# Defining data vector
```

```
x <- c(10, 11, 21, 22, 23, 25)

# output to be present as PNG file
png(file = "negativeskew.png")

# Print skewness of distribution
print(skewness(x))

# Histogram of distribution
hist(x)

# Saving the file
dev.off()
```

**Output:**

```
[1] -0.5794294
```

**Graphical Representation:**




*Simple Histogram*

Kurtosis
A statistical measure known as kurtosis measures the peakedness, flatness, and weight of the tails of data distributions. In a number of disciplines, including finance, economics, social sciences, and data analysis, an understanding of kurtosis is crucial. Kurtosis theory is thoroughly explained in this article, which also covers its definition, computation processes, interpretation, and applications.

Kurtosis is a numerical method in statistics that measures the sharpness of the peak in the data distribution.
**Formula:**
**where,**
*represents coefficient of kurtosis*
*represents mean of data vector*
***n** represents total number of observations*

There exist 3 types of Kurtosis values on the basis of which the sharpness of the peak is measured. These are as follows:

*Platykurtic*
Data distributions having flattened tails compared to the normal distribution are referred to statistically as platykurtic distributions. In several disciplines, including finance, economics, social sciences, and data analysis, it is essential to comprehend platykurtic data. The definition, calculation procedures, interpretation, and applications of Platykurtic.

If the coefficient of kurtosis is less than 3 i.e. , then the data distribution is platykurtic. Being platykurtic doesn't mean that the graph is flat-topped.
**Example:**
**R**

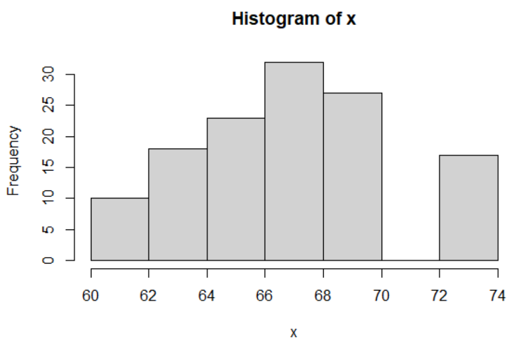
```
# Required for kurtosis() function

library(moments)



# Defining data vector

x <- c(rep(61, each = 10), rep(64, each = 18),

rep(65, each = 23), rep(67, each = 32), rep(70, each = 27),

rep(73, each = 17))
```

```
# output to be present as PNG file

png(file = "platykurtic.png")



# Print skewness of distribution

print(kurtosis(x))



# Histogram of distribution

hist(x)



# Saving the file

dev.off()
```

**Output:**

```
[1] 2.258318
```

**Graphical Representation:**




*Histogram*

*Mesokurtic*

Data distributions with tails that are similar in thickness to the normal distribution are known statistically as mesokurtic distributions. Numerous professions, including finance, economics, social sciences, and data analysis, depend on an understanding of mesokurtic data.

If the coefficient of kurtosis is equal to 3 or approximately close to 3 i.e. **gamma_{2}=3** , then the

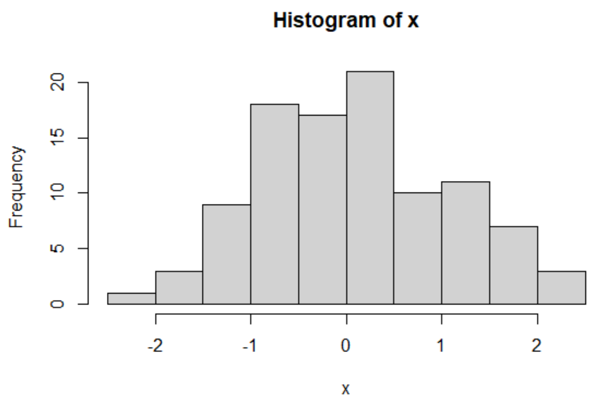
data distribution is mesokurtic. For the normal distribution, the kurtosis value is approximately equal to 3.
**Example:**
**R**

```
# Required for kurtosis() function

library(moments)



# Defining data vector

x <- rnorm(100)



# output to be present as PNG file

png(file = "mesokurtic.png")



# Print skewness of distribution

print(kurtosis(x))



# Histogram of distribution

hist(x)



# Saving the file

dev.off()
```

**Output:**

```
[1] 2.963836
```

**Graphical Representation:**




*Histogram*

*Leptokurtic*

Data distributions having hefty tails compared to the normal distribution are referred to statistically as leptokurtic distributions. In several disciplines, including finance, economics, social sciences, and data analysis, it is essential to comprehend leptokurtic data. The leptokurtic theory is thoroughly discussed in this article, including its concept, calculation techniques, interpretation, and applications.

If the coefficient of kurtosis is greater than 3 i.e.                    , then the data distribution is leptokurtic and shows a sharp peak on the graph.

**Example:**

**R**

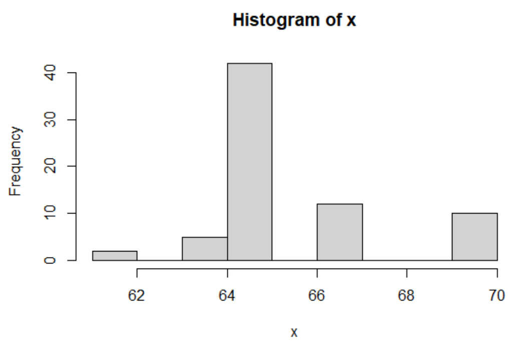
```
# Required for kurtosis() function

library(moments)



# Defining data vector

x <- c(rep(61, each = 2), rep(64, each = 5),

rep(65, each = 42), rep(67, each = 12), rep(70, each = 10))



# output to be present as PNG file
```

```
png(file = "leptokurtic.png")



# Print skewness of distribution

print(kurtosis(x))



# Histogram of distribution

hist(x)



# Saving the file

dev.off()
```

**Output:**

```
[1] 3.696788
```

**Graphical Representation:**




*Histogram*

## Box Plot

Box Plot is a graphical method to visualize data distribution for gaining insights and making informed decisions. Box plot is a type of chart that depicts a group of numerical data through their quartiles.

In this article, we are going to discuss **components of a box plot, how to create a box plot, uses of a Box Plot, and how to compare box plots.**

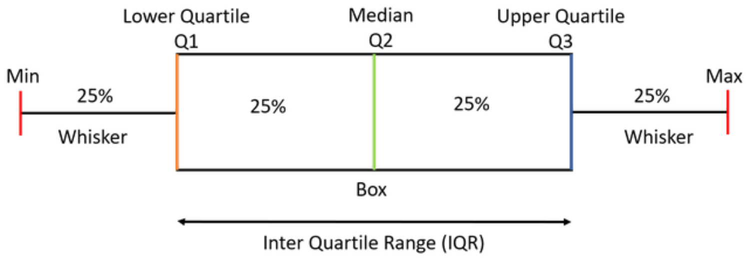
**What is a Box Plot?**

The idea of box plot was presented by John Tukey in 1970. He wrote about it in his book "Exploratory Data Analysis" in 1977. Box plot is also known as a whisker plot, box-and-whisker plot, or simply a box-and whisker diagram. Box plot is a graphical representation of the distribution of a dataset. It displays key summary statistics such as the median, quartiles, and potential outliers in a concise and visual manner. By using Box plot you can provide a summary of the distribution, identify potential and compare different datasets in a compact and visual manner.

Elements of Box Plot

A box plot gives a five-number summary of a set of data which is-

- **Minimum** – It is the minimum value in the dataset excluding the outliers.
- **First Quartile (Q1)** – 25% of the data lies below the First (lower) Quartile.
- **Median (Q2)** – It is the mid-point of the dataset. Half of the values lie below it and half above.
- **Third Quartile (Q3)** – 75% of the data lies below the Third (Upper) Quartile.
- **Maximum** – It is the maximum value in the dataset excluding the outliers.




The area inside the box (50% of the data) is known as the **Inter Quartile Range**. The **IQR** is calculated as –

IQR = Q3-Q1

**Outlies** are the data points **below and above** the **lower and upper limit**. The lower and upper limit is calculated as –

Lower Limit = Q1 - 1.5*IQR
Upper Limit = Q3 + 1.5*IQR

The values below and above these limits are considered outliers and the minimum and maximum values are calculated from the points which lie under the lower and upper limit.

## How to create a box plots?

Let us take a sample data to understand how to create a box plot.

Here are the runs scored by a cricket team in a league of 12 matches – ***100, 120, 110, 150, 110, 140, 130, 170, 120, 220, 140, 110.***

To draw a box plot for the given data first we need to arrange the data in ascending order and then find the minimum, first quartile, median, third quartile and the maximum.

**Ascending Order**
100, 110, 110, 110, 120, 120, 130, 140, 140, 150, 170, 220

**Median (Q2)** = (120+130)/2 = **125;** Since there were even values

To find the First Quartile we take the first six values and find their median.

**Q1** = (110+110)/2 = **110**

For the Third Quartile, we take the next six and find their median.

**Q3** = (140+150)/2 = **145**

**Note:** If the total number of values is odd then we exclude the Median while calculating Q1 and Q3. Here since there were two central values we included them. Now, we need to calculate the Inter Quartile Range.

**IQR** = Q3-Q1 = 145-110 = **35**

We can now calculate the Upper and Lower Limits to find the minimum and maximum values and also the outliers if any.

**Lower Limit** = Q1-1.5*IQR = 110-1.5*35 = **57.5**
**Upper Limit** = Q3+1.5*IQR = 145+1.5*35 = **197.5**

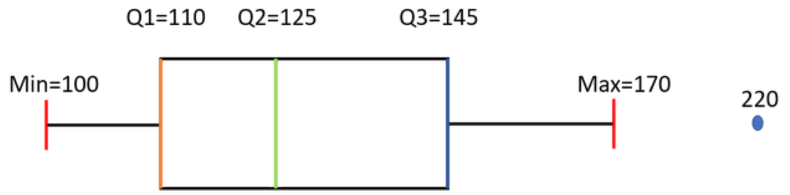
So, the minimum and maximum between the range [57.5,197.5] for our given data are –

**Minimum = 100**
**Maximum = 170**

The outliers which are outside this range are –

**Outliers = 220**

Now we have all the information, so we can draw the box plot which is as below-

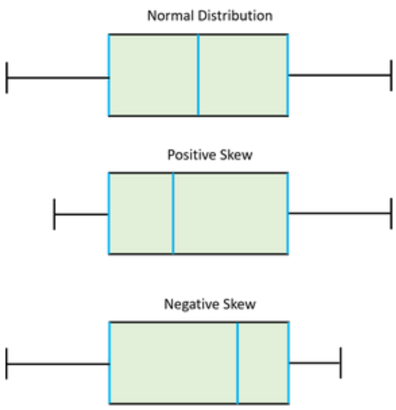



We can see from the diagram that the Median is not exactly at the center of the box and one whisker is longer than the other. We also have one Outlier.

## Use-Cases of Box Plot

- Box plots provide a visual summary of the data with which we can quickly identify the average value of the data, how dispersed the data is, whether the data is skewed or not (skewness).
- The Median gives you the average value of the data.
- Box Plots shows Skewness of the data-

**a)** If the Median is at the **center** of the Box and the **whiskers** are almost the **same on both the ends** then the data is **Normally Distributed**.
**b)** If the Median lies **closer to the First Quartile** and if the **whisker at the lower end is shorter** (as in the above example) then it has a **Positive Skew (Right Skew)**.
**c)** If the Median lies **closer to the Third Quartile** and if the **whisker at the upper end is shorter** than it has a **Negative Skew (Left Skew).**

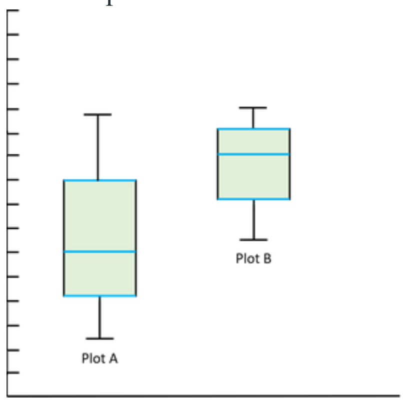



- The dispersion or spread of data can be visualized by the minimum and maximum values which are found at the end of the whiskers.
- The Box plot gives us the idea of about the Outliers which are the points which are numerically distant from the rest of the data.

**How to compare box plots?**

As we have discussed at the beginning of the article that box plots make comparing characteristics of data between categories very easy. Let us have a look at how we can compare different box plots and derive statistical conclusions from them.

Let us take the below two plots as an example: –




- **Compare the Medians** — If the median line of a box plot lies outside the box of the other box plot with which it is being compared, then we can say that there is likely to be a difference between the two groups. Here the Median line of the plot B lies outside the box of Plot A.
- **Compare the Dispersion or Spread of data** — The Inter Quartile range (length of the box) gives us an idea about how dispersed the data is. Here Plot A has a longer length than Plot B which means that the dispersion of data is more in plot A as compared to plot B. The length of whiskers also gives an idea of the overall spread of data. The extreme values (minimum &maximum) give the range of data distribution. Larger the range more scattered the data. Here Plot A has a larger range than Plot B.
- **Comparing Outliers** — The outliers give the idea of unusual data values which are distant from the rest of the data. More number of Outliers means the prediction will be more uncertain. We can be more confident while predicting the values for a plot which has less or no outliers.
- **Compare Skewness** — Skewness gives us the direction and the magnitude of the lack of symmetry. We have discussed above how to identify skewness. Here Plot A is Positive or Right Skewed and Plot B is Negative or Left Skewed.

This is all for Box Plots. Now you might have got the idea of Box Plots how to make them and how to derive information from them. For any queries do leave a comment down below.

**Box plot – Frequently Asked Questions (FAQs)**

What do you mean by box plot?
*A box plot, also known as a box-and-whisker plot, is a graphical representation of the distribution of a dataset. It summarizes key statistics such as the median, quartiles, and outliers, providing insights into the spread and central tendency of the data.*
*Box Plot is used for which type of data?*
*Box Plots gives a visual summary of the variability of values of dataset. Boxplots usually shows the numeric data values; especially is you want to compare multiple groups.*
*What information cannot be found in a box plot?*
*Information that are missed in a box plot is the detailed shape of the distribution. It is quite difficult to find the mean as it is visual representation of the data.*
*Is Box Plot vertical or horizontal?*
*Box Plot can either be drawn horizontally or vertically. It depends on the estimate L-estimators, range, mid-range and trimean.*

**Pivot Tables in Pandas**
In this article, we will see the Pivot Tables in Pandas. Let's discuss some concepts:

**Pandas:** Pandas is an open-source library that is built on top of the NumPy library. It is a Python package that offers various data structures and operations for manipulating numerical data and time series. It is mainly popular for importing and analysing data much easier. Pandas is fast and it has high-performance & productivity for users.
**Pivot Tables:** A pivot table is a table of statistics that summarizes the data of a more extensive table (such as from a database, spreadsheet, or business intelligence program). This summary might include sums, averages, or other statistics, which the pivot table groups together in a meaningful way.

Steps Needed

- Import Library (Pandas)
- Import / Load / Create data.
- Use Pandas.pivot_table() method with different variants.

Here, we will discuss some variants of pivot table over the dataframe shown below :

- Python3

```
# import packages

import pandas as pd



# create data

df = pd.DataFrame({'ID': {0: 23, 1: 43, 2: 12,
```

```
                            3: 13, 4: 67, 5: 89,

                            6: 90, 7: 56, 8: 34},



                 'Name': {0: 'Ram', 1: 'Deep', 2: 'Yash',

                            3: 'Aman', 4: 'Arjun', 5: 'Aditya',

                            6: 'Akash', 7: 'Chalsea',

                            8: 'Divya'},



                 'Marks': {0: 89, 1: 97, 2: 45,

                             3: 78, 4: 56, 5: 76,

                             6: 81, 7: 87, 8: 100},



                 'Grade': {0: 'B', 1: 'A', 2: 'F',

                             3: 'C', 4: 'E', 5: 'C',

                             6: 'B', 7: 'B', 8: 'A'}})



# view data

display(df)
```

**Output:**

| | ID | Name | Marks | Grade |
|---|---|---|---|---|
| 0 | 23 | Ram | 89 | B |
| 1 | 43 | Deep | 97 | A |
| 2 | 12 | Yash | 45 | F |
| 3 | 13 | Aman | 78 | C |
| 4 | 67 | Arjun | 56 | E |
| 5 | 89 | Aditya | 76 | C |
| 6 | 90 | Akash | 81 | B |
| 7 | 56 | Chalsea | 87 | B |
| 8 | 34 | Divya | 100 | A |

### Example 1: Simple use of pivot_table() method.

- Python3

```
# import packages

import pandas as pd



# create data

df = pd.DataFrame({'ID': {0: 23, 1: 43, 2: 12,

                          3: 13, 4: 67, 5: 89,

                          6: 90, 7: 56, 8: 34},



                   'Name': {0: 'Ram', 1: 'Deep', 2: 'Yash',

                            3: 'Aman', 4: 'Arjun', 5: 'Aditya',

                            6: 'Akash', 7: 'Chalsea',
```

```
                          8: 'Divya'},



                 'Marks': {0: 89, 1: 97, 2: 45,

                           3: 78, 4: 56, 5: 76,

                           6: 81, 7: 87, 8: 100},



                 'Grade': {0: 'B', 1: 'A', 2: 'F',

                           3: 'C', 4: 'E', 5: 'C',

                           6: 'B', 7: 'B', 8: 'A'}})



# simple use pivot_table() method

print(pd.pivot_table(df, index = ["ID"]))
```

**Output :**

| | Marks |
|---|---|
| **ID** | |
| 12 | 45 |
| 13 | 78 |
| 23 | 89 |
| 34 | 100 |
| 43 | 97 |
| 56 | 87 |
| 67 | 56 |
| 89 | 76 |
| 90 | 81 |

**Example 2: Pivot table with multiple columns indexes.**

- Python3

```
# import packages

import pandas as pd



# create data

df = pd.DataFrame({'ID': {0: 23, 1: 43, 2: 12,

                          3: 13, 4: 67, 5: 89,

                          6: 90, 7: 56, 8: 34},



                 'Name': {0: 'Ram', 1: 'Deep', 2: 'Yash',

                          3: 'Aman', 4: 'Arjun', 5: 'Aditya',

                          6: 'Akash', 7: 'Chalsea',

                          8: 'Divya'},



                 'Marks': {0: 89, 1: 97, 2: 45,

                           3: 78, 4: 56, 5: 76,

                           6: 81, 7: 87, 8: 100},



                 'Grade': {0: 'B', 1: 'A', 2: 'F',

                           3: 'C', 4: 'E', 5: 'C',

                           6: 'B', 7: 'B', 8: 'A'}})

# multiple columns with
```

```
# pivot_table() method

display(pd.pivot_table(df,

                       index = ["ID", "Name"]))
```

## Output :

| | | Marks |
|---|---|---|
| **ID** | **Name** | |
| 12 | Yash | 45 |
| 13 | Aman | 78 |
| 23 | Ram | 89 |
| 34 | Divya | 100 |
| 43 | Deep | 97 |
| 56 | Chalsea | 87 |
| 67 | Arjun | 56 |
| 89 | Aditya | 76 |
| 90 | Akash | 81 |

## Example 3: Pivot table with an aggregate function.

- Python3

```
# import packages

import pandas as pd

import numpy as np



# create data

df = pd.DataFrame({'ID': {0: 23, 1: 43, 2: 12,

                          3: 13, 4: 67, 5: 89,

                          6: 90, 7: 56, 8: 34},
```

```
                    'Name': {0: 'Ram', 1: 'Deep',
                             2: 'Yash', 3: 'Aman',
                             4: 'Arjun', 5: 'Aditya',
                             6: 'Akash',7: 'Chalsea',
                             8: 'Divya'},

                    'Marks': {0: 89, 1: 97, 2: 45,
                              3: 78, 4: 56, 5: 76,
                              6: 81, 7: 87, 8: 100},

                    'Grade': {0: 'B', 1: 'A', 2: 'F', 3: 'C',
                              4: 'E', 5: 'C', 6: 'B', 7: 'B',
                              8: 'A'}})

# Pivot Table with mean
# aggregate function on marks
display(pd.pivot_table(df,
                     index = ["Grade"],
                     values = ["Marks"],
                     aggfunc = np.mean))
```

**Output :**

| | Marks |
|---|---|
| Grade | |
| A | 98.500000 |
| B | 85.666667 |
| C | 77.000000 |
| E | 56.000000 |
| F | 45.000000 |

# What is Heat map Data Visualization and How to Use It?

Heatmap data visualization is a powerful tool used to represent numerical data graphically, where values are depicted using colors. This method is particularly effective for identifying patterns, trends, and anomalies within large datasets. This article will explore what heatmap data visualization is, its types, benefits, and best practices for using it effectively.

## What is Heatmap Data Visualization?

Heatmap data visualization is a technique that uses color to represent data values. The most common color schemes range from warm colors (such as red) to cool colors (such as blue), with warm colors typically representing higher values and cool colors representing lower values. This visual representation allows for quick and intuitive understanding of complex data sets.

At its core, a heatmap is a graphical representation of data where values are depicted using colors. The data is typically arranged in a grid or matrix format, with each cell assigned a color based on its value. The intensity of the color corresponds to the magnitude of the data, allowing viewers to discern patterns and trends at a glance. Heatmaps are particularly useful for visualizing large datasets and identifying areas of interest or concentration.

## Why Heatmap Data Visualization?

Heatmaps offer a powerful way to visualize and analyze data, providing several advantages that make them a popular choice for data visualization:

- **Identifying Patterns and Trends:** Heatmaps allow for quick identification of patterns, trends, and correlations within datasets. By visually representing data with color gradients, heatmaps make it easier to discern areas of high or low concentration, enabling analysts to uncover insights that might not be apparent from raw data alone.
- **Handling Large Datasets:** Heatmaps are particularly effective for visualizing large datasets. They condense complex information into a compact, easily interpretable format, making it feasible to analyze and extract meaningful information from extensive data sets.
- **Intuitive Interpretation:** Heatmaps provide a visual representation of data that is intuitive and easy to interpret. The color gradients used in heatmaps offer a straightforward way to understand data density, intensity, and distribution, allowing viewers to grasp trends and patterns at a glance.
- **Comparing Data Sets**: Heatmaps enable side-by-side comparison of multiple datasets, facilitating the identification of similarities, differences, and relationships between variables. By overlaying or displaying multiple heatmaps simultaneously, analysts can gain insights into how different factors interact and influence each other.
- **Facilitating Decision Making:** Heatmaps help decision-makers make informed decisions by presenting complex data in a visually compelling format. Whether it's identifying areas of opportunity, allocating resources, or optimizing processes, heatmaps provide a clear and concise way to communicate insights and drive action.
- **Versatility:** Heatmaps can be applied across various domains and disciplines, including finance, healthcare, marketing, and science. From analyzing financial trends to tracking user behavior on websites, heatmaps offer a versatile tool for data visualization and analysis in diverse contexts.
- **Enhancing Communication:** Heatmaps are effective tools for communicating insights and findings to stakeholders, colleagues, and clients. Whether it's presenting research findings,

reporting on key performance indicators, or visualizing spatial data, heatmaps help convey complex information in a visually appealing and accessible manner.

## Types of Heatmaps

Heatmaps can be categorized based on the type of data they visualize and the specific use case. Here are some common types:

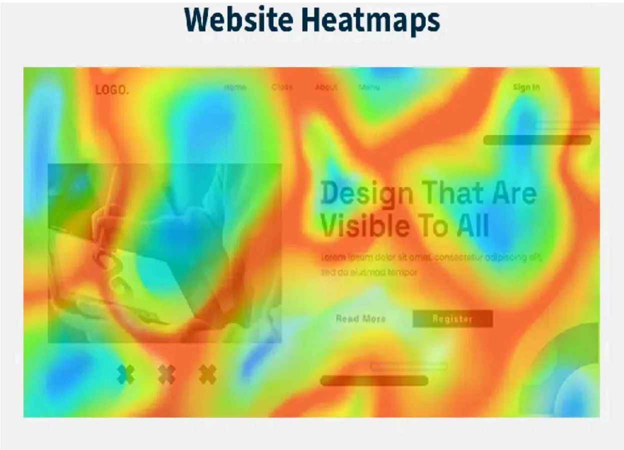
1. Website Heatmaps

Imagine you have a website, and you want to understand how visitors interact with it. A heatmap is like a map that shows you where visitors are spending the most time and where they're not. Think of it like this: the more time visitors spend on a particular section of your site, the "hotter" it gets on the heatmap. This is usually shown with warm colors like red or orange. So, if a section is red, it means it's getting a lot of attention.

Conversely, if a section is blue or green, it's "cooler," meaning visitors aren't spending much time there. So, blue or green areas indicate lower interaction.

Website heatmaps are used to visualize user behavior on web pages. They help identify which parts of a webpage receive the most interaction, such as clicks, scrolls, and mouse movements.

- **Click Maps**: Show where users click on a webpage, helping to identify popular links and buttons.
- **Scroll Maps**: Indicate how far users scroll down a page, revealing which sections are most engaging.
- **Mouse Tracking Heatmaps**: Track mouse movements to understand which areas of a page attract the most attention.
- **Eye-Tracking Heatmaps**: Visualize where users' eyes focus on a page, providing insights into visual engagement.



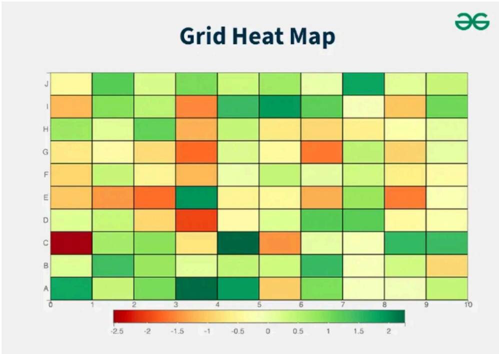

2. Grid Heatmaps

Grid heatmaps are a powerful visualization tool used to represent data in a tabular format where each cell's color indicates the value of the data point it represents. This method is particularly effective for comparing multiple variables and identifying patterns, trends, and correlations within the data.

**Key Characteristics:**

1. **Color Encoding**: The primary feature of a heatmap is its use of color to represent data values. Colors are typically chosen from a gradient, with the intensity or hue indicating the magnitude of the values. For example, in a common color scheme, darker colors might represent higher values, while lighter colors represent lower values.
2. **Grid Layout**: The data is displayed in a grid, where each cell corresponds to a specific data point. The position of each cell is determined by its row and column, which often correspond to specific variables or categories.
3. **Data Density**: Heatmaps can handle large datasets effectively, making it easy to spot anomalies, trends, and clusters within the data. This is particularly useful in fields such as biology, where gene expression levels across different conditions can be visualized, or in finance, where stock price movements across different time periods are compared.

4. **Comparative Analysis**: By visualizing data in a grid format, heatmaps facilitate the comparison of multiple variables simultaneously. This can help identify correlations or relationships between different data points.



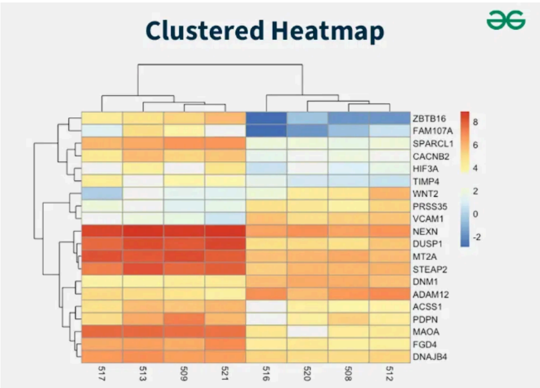

*Grid Heatmaps*

3. Clustered Heatmaps

Clustered heatmaps extend the functionality of standard grid heatmaps by incorporating hierarchical clustering to show relationships between rows and columns. This added dimension of information makes clustered heatmaps particularly valuable in fields like biology, where they are commonly used to visualize genetic data.

**Key Characteristics:**

1. **Hierarchical Clustering**: Clustered heatmaps use hierarchical clustering algorithms to group similar rows and columns together. This clustering is often displayed as dendrograms (tree-like diagrams) alongside the heatmap, indicating the similarity between different rows or columns.
2. **Color Encoding**: As with standard heatmaps, the cell color represents the value of the data point. The color intensity or hue typically indicates the magnitude of the values, allowing for easy visual differentiation.
3. **Enhanced Patterns and Relationships**: By clustering similar rows and columns together, clustered heatmaps make it easier to identify patterns, correlations, and relationships within the data. This can reveal underlying structures that might not be immediately apparent in a standard heatmap.
4. **Interactive Exploration**: Many software tools and libraries allow users to interact with clustered heatmaps, enabling them to zoom in on specific clusters, reorder rows and columns, and explore the data in greater detail.




*Clustered Heatmaps*

**Benefits of Heatmap Visualization**

Heatmaps offer several advantages over traditional data visualization methods:

- **Intuitive Understanding**: Colors make it easy to grasp complex data at a glance.

- **Pattern Recognition**: Heatmaps help identify patterns and trends that might be missed in numerical data.
- **Engagement**: The use of color makes heatmaps visually appealing and engaging.
- **Granularity**: Heatmaps provide detailed insights into data, allowing for more granular analysis.

**When to Use Heatmap Visualization**

Heatmaps are versatile and can be used in various scenarios:

- **Website Optimization**: To understand user behavior and optimize webpage design.
- **Financial Analysis**: To visualize performance metrics and identify areas needing improvement.
- **Marketing**: To track campaign performance and customer engagement.
- **Scientific Research**: To analyze genetic data and other complex datasets.
- **Geographic Analysis**: Visualizing spatial data such as population density, crime rates, or weather patterns.
- **Sports Analytics**: Analyzing player movements, game strategies, or performance metrics.

**Best Practices for Using Heatmaps for Data Visualization**

To effectively use heatmaps, consider the following best practices:

- **Choose the Right Color Scale**: Selecting an appropriate color scale is crucial. Sequential scales are ideal for data that progresses in one direction, while diverging scales are suitable for data with a central neutral point and values that can be both positive and negative.
- **Ensure Sufficient Data:** Heatmaps require a large amount of data to be accurate. Analyzing heatmaps with insufficient data can lead to misleading conclusions.
- **Combine with Other Analytics**: Heatmaps should be used in conjunction with other analytics tools to provide a comprehensive understanding of the data. For example, combining heatmaps with form analytics can offer deeper insights into user behavior.
- **Use Legends**: Always include a legend to help interpret the color scale used in the heatmap. This ensures that viewers can accurately understand the data being presented.
- **Highlight Key Areas**: Use heatmaps to draw attention to important areas of the data. For example, in a website heatmap, highlight areas with the most user interaction to focus on optimizing those sections.

**Different Tools for Generating Heatmaps**

When it comes to generating heatmaps, several tools stand out for their features, ease of use, and effectiveness. Here are some of the best tools for generating heatmaps:

1. **VWO Insights:** VWO Insights is a comprehensive tool that offers advanced heatmap features. It provides predictive attention heatmaps, click maps, and scroll maps. VWO Insights is particularly useful for identifying user bottlenecks and optimizing website design. It integrates well with other tools for a holistic approach to user experience and conversion rate optimization.
2. **Hotjar:** Hotjar is a popular tool for creating website heatmaps. It supports click maps, scroll maps, and movement heatmaps. Hotjar also offers additional features like session recordings, feedback, surveys, and interviews, making it a robust tool for understanding user behavior and improving user experience. The free plan allows up to 35 daily sessions, and there are paid plans with more advanced features.
3. **SmartLook:** SmartLook provides three types of heatmaps: click maps, movement heatmaps, and scroll maps. It also offers session recordings, events, funnels, and crash reports. SmartLook is known for its cross-device viewing capabilities and integration with third-party testing solutions. However, the free version has limitations on sharing and downloading heatmaps.
4. **Microsoft Clarity:** Microsoft Clarity is a free tool that offers heatmaps along with session recordings and other analytics features. It is designed to help users understand how visitors interact with their website and identify areas for improvement.
5. **Google Analytics (Page Analytics):** Google Analytics offers a heatmap feature through its Chrome extension, Page Analytics. This tool provides a visual representation of where visitors click on a webpage, helping to identify popular and underperforming elements.

**Conclusion**

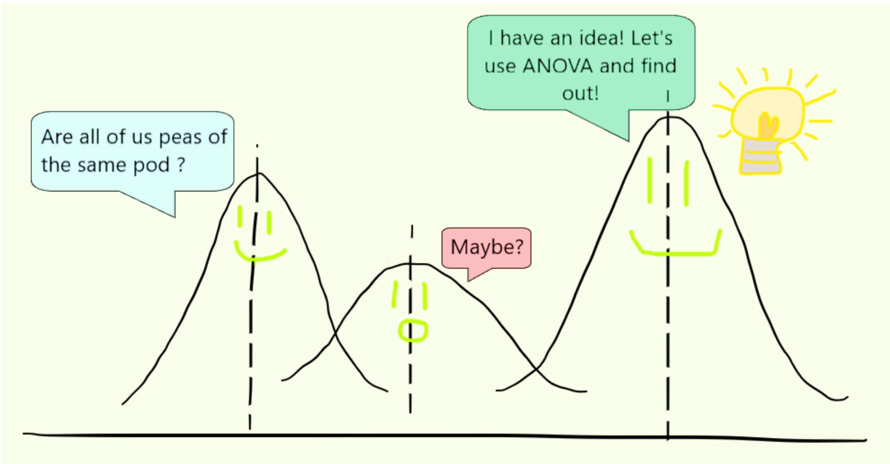
Heatmap data visualization is a powerful tool for representing complex data in an intuitive and engaging way. By using color to depict data values, heatmaps make it easier to identify patterns, trends, and anomalies. Whether used for website optimization, financial analysis, or scientific research, heatmaps provide valuable insights that can drive better decision-making.

# One-Way ANOVA

ANOVA, or Analysis of Variance, is a statistical method for comparing means among three or more groups, crucial in understanding group differences and relationships in diverse fields. In this article, we'll focus on One-way ANOVA.

## What is ANOVA?

ANOVA, or Analysis of Variance is a parametric statistical technique that helps in finding out if there is a significant difference between the mean of three or more groups. It checks the impact of various factors by comparing groups (samples) based on their respective mean. ANOVA tests the null hypothesis that all group means are equal, against the alternative hypothesis that at least one group mean is different.




# Assumptions for ANOVA

1. The dependent variable is approximately normally distributed within each group. This assumption is more critical for smaller sample sizes.
2. The samples are selected at random and should be independent of one another.
3. All groups have equal standard deviations.
4. Each data point should belong to one and only one group. There should be no overlap or sharing of data points between groups.

In the case of two-way ANOVA, there are additional assumptions related to the interaction between the independent variables.

- The effect of one independent variable on the dependent variable should be consistent across all levels of the other independent variable.
- Combined effect of two independent variables is equal to the sum of their individual effects.

**Types of ANOVA**

There are two main types of ANOVA:

1. **One-way ANOVA**: This is the most basic form of ANOVA and is used when there is only one independent variable with more than two levels or groups. It assesses whether there are any statistically significant differences among the means of the groups.
2. **Two-way ANOVA**: Extending the one-way ANOVA, two-way ANOVA involves two independent variables. It allows for the examination of the main effects of each variable as well as the interaction between them. The interaction effect explores whether the effect of one variable on the dependent variable is different depending on the level of the other variable.

**ANOVA (Analysis of Variance) Test in R Programming**

ANOVA also known as Analysis of variance is used to investigate relations between categorical variables and continuous variables in the R Programming Language. It is a type of hypothesis testing for population variance. It enables us to assess whether observed variations in means are statistically significant or merely the result of chance by comparing the variation within groups to the variation between groups. The ANOVA test is frequently used in many disciplines, including business, social sciences, biology, and experimental research.

R – ANOVA Test

ANOVA tests may be run in R programming, and there are a number of functions and packages available to do so.

ANOVA test involves setting up:

- **Null Hypothesis:** The default assumption, or null hypothesis, is that there is no meaningful relationship or impact between the variables. It stands for the absence of a population-wide link, difference, or effect. The statement that two or more groups are equal or that the effect size is zero is sometimes expressed as the null hypothesis. The null hypothesis is commonly written as H0.

- **Alternate Hypothesis:** The opposite of the null hypothesis is the alternative hypothesis. It implies that there is a significant relationship, difference, or link among the population's variables. Depending on the study question or the nature of the issue under investigation, it may take several forms. Alternative hypotheses are sometimes referred to as H1 or HA.

**ANOVA tests are of two types:**

- **One-way ANOVA:** One-way When there is a single categorical independent variable (also known as a factor) and a single continuous dependent variable, an ANOVA is employed. It seeks to ascertain whether there are any notable variations in the dependent variable's means across the levels of the independent variable.

- **Two-way ANOVA:** When there are two categorical independent variables (factors) and one continuous dependent variable, two-way ANOVA is used as an extension of one-way ANOVA. You can evaluate both the direct impacts of each independent variable and how they interact with one another on the dependent variable.

The Dataset

The mtcars(motor trend car road test) dataset is used which consist of 32 car brands and 11 attributes. The dataset comes preinstalled in **dplyr** package in R.

To get started with ANOVA, we need to install and load the **dplyr** package.

Performing One Way ANOVA test in R language

One-way ANOVA test is performed using mtcars dataset which comes preinstalled with dplyr package between disp attribute, a continuous attribute and gear attribute, a categorical attribute.here are some steps.

- Setup Null Hypothesis and Alternate Hypothesis
- H0 = mu = mu01 = mu02(There is no differencebetween average displacement for different gears)
- H1 = Not all means are equal.

- R

```
# Installing the package

install.packages("dplyr")
```

```
# Loading the package

library(dplyr)



head(mtcars)
```

**Output:**

```
mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

Here we will print top 5 record of our dataset to get an idea about our dataset.
Perform the ANOVA test using aov function.

- R

```
mtcars_aov <- aov(mtcars$disp~factor(mtcars$gear))

summary(mtcars_aov)
```

**Output:**

```
Df Sum Sq Mean Sq F value   Pr(>F)
factor(mtcars$gear)  2 280221  140110   20.73 2.56e-06 ***
Residuals           29 195964    6757
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- **Df:** The model’s degrees of freedom.
- **Sum Sq:** The sums of squares, which represent the variability that the model is able to account for.
- **Mean Sq:** The variance explained by each component is represented by the mean squares.
- **F-value:** It is the measure used to compare the mean squares both within and between groups.
- **Pr(>F):** The F-statistics p-value, which denotes the factors’ statistical significance.
- **Residuals:** Relative deviations from the group mean, are often known as residuals and their summary statistics.

Identifier codes: Asterisks (*) are used to show the degree of significance; they stand for p 0.05, p 0.01, and p 0.001, respectively.

Performing Two Way ANOVA test in R

A two-way ANOVA test is performed using mtcars dataset which comes preinstalled with dplyr package between disp attribute, a continuous attribute and gear attribute, a categorical attribute, am attribute, a categorical attribute.

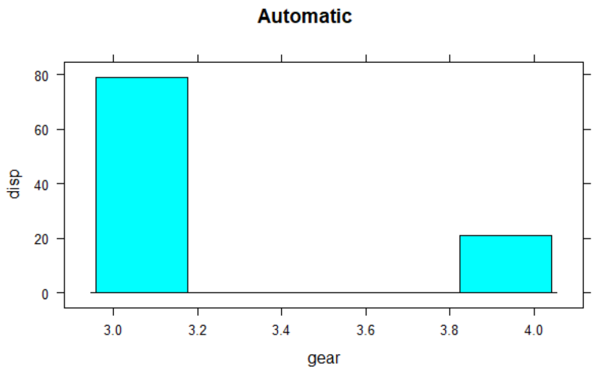
- Setup Null Hypothesis and Alternate Hypothesis

- H0 = mu0 = mu01 = mu02(There is no difference between average displacement for different gear)
- H1 = Not all means are equal

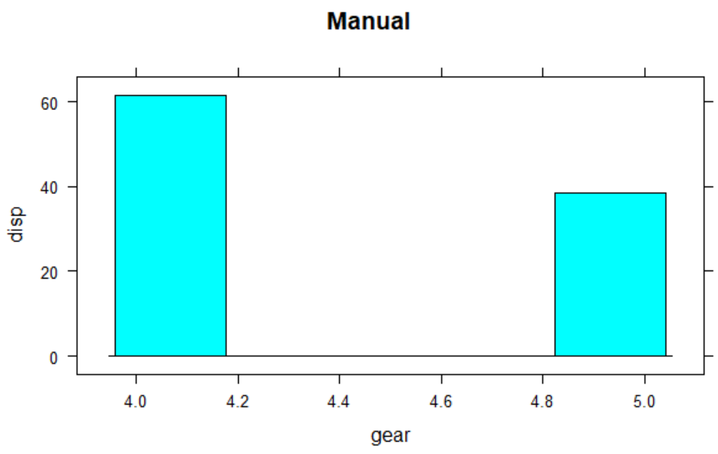
- R

```
# Installing the package

install.packages("dplyr")



# Loading the package

library(dplyr)



# Variance in mean within group and between group

histogram(mtcars$disp~mtcars$gear, subset = (mtcars$am == 0),

xlab = "gear", ylab = "disp", main = "Automatic")

histogram(mtcars$disp~mtcars$gear, subset = (mtcars$am == 1),

xlab = "gear", ylab = "disp", main = "Manual")
```

**Output:**




*ANOVA Test in R Programming*




*ANOVA Test in R Programming*

The histogram shows the mean values of gear with respect to displacement. Hear categorical variables are gear and am on which factor function is used and continuous variable is disp.
Calculate test statistics using aov function

- R

```
mtcars_aov2 <- aov(mtcars$disp~factor(mtcars$gear) *

factor(mtcars$am))

summary(mtcars_aov2)
```

**Output:**

```
Df Sum Sq Mean Sq F value   Pr(>F)
factor(mtcars$gear)  2 280221  140110  20.695 3.03e-06 ***
factor(mtcars$am)    1   6399    6399   0.945    0.339
Residuals           28 189565    6770
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

The summary shows that the gear attribute is very significant to displacement(Three stars denoting it) and am attribute is not much significant to displacement. P-value of gear is less than 0.05, so it proves that gear is significant to displacement i.e related to each other. P-value of am is greater than 0.05, am is not significant to displacement i.e not related to each other.
Find the best-fit model
We have two different anova models and we will try to find the best fit model based on their AIC score.
The Akaike Information Criterion (AIC), which accounts for the number of predictors, is a gauge of a model’s goodness of fit. It penalizes more intricate models in order to prevent overfitting. Better-fitting models are indicated by lower AIC values.

- R

```
library(AICcmodavg)

model.set <- list(mtcars_aov, mtcars_aov2)

model.names <- c("mtcars_aov", "mtcars_aov2")

aictab(model.set, modnames = model.names)
```

**Output:**

```
Model selection based on AICc:

            K   AICc Delta_AICc AICcWt Cum.Wt      LL
mtcars_aov  4 379.33       0.00   0.71   0.71 -184.93
mtcars_aov2 5 381.10       1.76   0.29   1.00 -184.39
```

- **AICc (Corrected AIC):** AICc is a measure of how well a statistical model fits the data. Lower AICc values indicate better-fitting models.
- **Delta_AICc (Difference in AICc):** This column represents the difference in AICc between each model and the best-fitting model. Smaller values are better, and a difference of 2 or more is considered significant.
- **AICcWt (AICc Weight):** AICc weight indicates the probability that a given model is the best among the ones considered. In your table, the model with the highest AICc weight (0.71) is considered the most likely best model.
- **Cum.Wt (Cumulative AICc Weight):** This shows the cumulative probability that any model up to a particular row is the best-fitting model.
- **LL (Log-Likelihood):** Log-likelihood measures how well a model explains the observed data. Higher values mean a better fit.

Plot the results in a graph

We will plot both the model together so find out the compression of both the models. here we will use ggplot library and plot the box plot and visualize our both models.

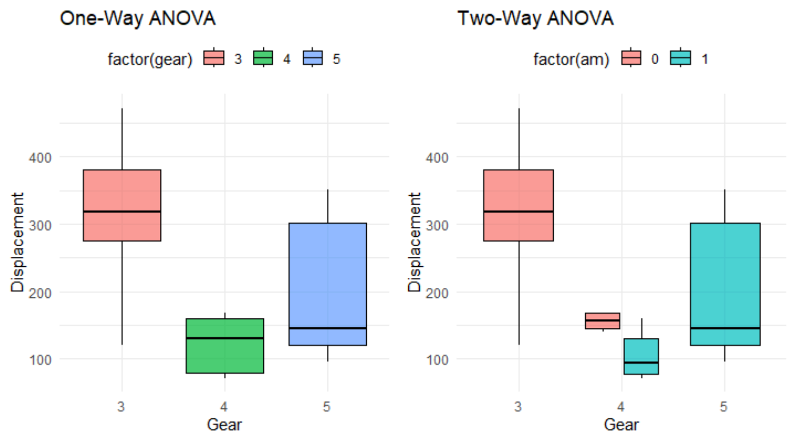
- R

```
# Load required packages

install.packages("ggplot2")

library(ggplot2)
```

```
# One-way ANOVA visualization

plot1 <- ggplot(mtcars, aes(x = factor(gear), y = disp, fill = factor(gear))) +

geom_boxplot(color = "black", alpha = 0.7) +

labs(title = "One-Way ANOVA",

x = "Gear",

y = "Displacement") +

theme_minimal() +

theme(legend.position = "top")



# Two-way ANOVA visualization

plot2 <- ggplot(mtcars, aes(x = factor(gear), y = disp, fill = factor(am))) +

geom_boxplot(color = "black", alpha = 0.7) +

labs(title = "Two-Way ANOVA",

x = "Gear",

y = "Displacement") +

theme_minimal() +

theme(legend.position = "top")



# Combine the plots for comparison

library(gridExtra)
```

```
grid.arrange(plot1, plot2, ncol = 2)
```

**Output:**




*Anova Test In R*

The box plots visually compare the displacement (disp) distribution across different gear levels for both one-way and two-way ANOVA models. In the one-way ANOVA, each box represents a gear level, showcasing the variability in displacements.

The two-way ANOVA extends this comparison, incorporating the additional factor (am), providing a more detailed insight into how both factors collectively influence displacement. The plots help discern any notable differences or patterns in dispersion, aiding in the interpretation of model effects on the response variable.